

Gravity-Light

Universal Quantum Foam Hypothesis

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Abstract

This work formulates the Universal Quantum Foam Hypothesis (UQSH) as an ontological field framework. Its starting point is the Qu-ground Q , whose physically readable field projection Φ appears as Qu-foam. The motion of the Qu-foam field $B(\Phi)$ forms the common basis of light, energy, and gravitational field action:

$$Q \rightarrow \Phi \rightarrow B(\Phi), \quad B(\Phi) \Rightarrow \{L, E, \mathfrak{G}\}.$$

Within this framework, light is not understood as a flying object, but as a free, spherically propagating field motion. Energy describes the motion-content of this dynamics, while gravitation describes its regime-effective curvature and tension aspect. Matter arises as bound field motion, and inertia as the resistance of this binding to forced reorganization.

The work interprets redshift as the dedensification of propagating tension fronts, the dark-matter effect as baryonically anchored field tension, and the dark-energy effect as large-scale relaxation and dedensification dynamics. It further addresses quanta, spectrum, uncertainty, Anektron and Anun, magnetism, chemistry, space, time, the speed of light, matter formation, and the Planck limit as a regime-dependent boundary of visibility.

Hypothesis Status and Scale-Relative Working Level

The following sections should be understood as a partially formalized UQSH hypothesis. The UQSH does not replace established measurements or successful mathematical relations in physics. Quantities such as the speed of light, spectral lines, the energy–mass relation, gravitational approximations, rotation curves, lensing signals, or quantized coupling events remain reference points against which every UQSH interpretation must be tested.

The aim of this work is therefore not to reject confirmed formalisms, but to extend their ontological reading. Within the UQSH framework, established equations can be understood as limiting cases, regime projections, or effective coupling forms of a deeper field organization. Proximity to known measured quantities is explicitly intended: a UQSH interpretation is viable only if it remains connectable to existing mathematical and physical descriptions.

Within this work, the scale-relative basic assumption of the UQSH is taken as the supporting working level: the physically readable field reality organizes itself across different scales according to recurring principles of saturation, coupling, coherence, inertia, reorganization, and boundary formation. This recurrence, however, does not imply a literal identification of different objects.

Terms such as electron-like, proton-like, neutron-like, atom-like, or element-analogous therefore denote scale-relative functional roles within the same field logic. A galaxy is not claimed to be an actual atom. In a superregime, however, it may possess an atom-like function when central condensation, peripheral coupling, neutral stabilization, and transitional modes develop the same organizational pattern.

The UQSH thus treats atoms, galaxies, and cosmic structures as different regimes of a recurring field-working mode. What matters is not external similarity, but the recurrence of roles: center, binding space, shell, neutralization, coupling, saturation, reorganization, and transition.

All statements in this section should therefore be understood as working hypotheses. They gain strength not through the mere assertion of an analogy, but only through compatibility with observational data, formal refinement, and the examination of whether the same organizational roles actually recur across different scales.

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1. Introduction

The present work formulates the Universal Quantum Foam Hypothesis (UQSH) as an ontological field framework. Its starting point is not an individual particle, not light, not matter, and not gravitation, but the Qu-foam itself. In what follows, this is referred to as the *Qu-ground* (Q). This denotes the deepest basis from which physical field reality, motion, curvature, coupling, matter, space, time, and observation become readable at all.

In order to distinguish this fundamental level from physically accessible field reality, the notation (Φ) will later be introduced for the field projection of the Qu-ground. The term *projection* does not mean that the physical world would be merely virtual, apparent, or less real. It is not meant as a mere image without reality, nor as a simulation. Rather, (Φ) denotes the physically real field form in which the Qu-ground becomes readable for a regime as field medium, motion, tension, curvature, and organization.

How (Q) itself is connected to this physical field reality cannot, according to the present state of knowledge in this work, be measured directly. What is measurable is not the Qu-ground as such, but its regime-dependent field effects: radiation, matter, gravitation, spectrum, coupling, saturation, dedensification, and reorganization.

Within the UQSH framework, the search for particles behind the dark-matter effect takes on a different meaning. It is not wrong, but it may be directed toward an ontological expectation that is too narrow. What is being searched for may not be an additional substance in space, but a previously overlooked part of physical reality: the field medium itself. In the standard picture, this medium appears only indirectly as space, geometry, or background. In the UQSH, by contrast, it is the active carrier of matter, radiation, gravitation, and observation. The dark-matter effect would then not be the signal of a hidden particle, but the measurable effect of the very field that first determines what can appear as particle, mass, light, or gravitation.

In established physics, light is successfully described as an electromagnetic wave, a photon, or a quantized field excitation. These descriptions are precise and very well confirmed experimentally. Yet a deeper ontological question remains open: What is actually propagating when light travels from a source to an observer? Is it a stream of particles, a wave of a field, a change in spacetime, or a combination of different descriptions?

The UQSH answers this question field-mechanically: light is not a substance flying through empty space. It is an ongoing reorganization of the field medium itself. Put more simply, light is a form of motion of the universe. A helpful image is a wave spreading in circles across calm water after a stone has been thrown into it. The stone releases a tension, the water reorganizes, and the disturbance continues as a wave. Within the UQSH framework, something comparable happens in the field medium: a local excitation produces a tension front that propagates not merely circularly across a surface, but spherically through the surrounding field space. If a tension were released in the middle of water, its natural propagation there too would be spatial and spherical.

In this work, the term *gravitational light* denotes precisely this deeper field interpretation. It does not mean that ordinary light is simply gravitation in the classical sense. Rather, it means that light is understood as a free, spherically propagating tension front of the same field medium whose organized curvature also appears as gravitation. Light therefore carries not only electromagnetically readable modulations such as frequency, color, or spectrum. It is also real field motion: tension, timing,

reorganization, and, in the limiting case, also curvature action of the same gravity-capable field.

Radiation of different strengths then does not describe a fundamentally different kind of motion, but different degrees of tension and timing within the same Qu-foam motion. Within the UQSH framework, radio waves, visible light, X-rays, and gamma radiation differ above all in how densely, strongly, and rapidly they reorganize the field.

Here, too, the image of water and stone is useful: a small stone produces a gentle wave pattern with widely spaced waves; a larger or faster impacting stone produces denser, stronger, and more energetic waves. The water remains the same medium, but the released tension, timing, and wave strength differ. Likewise, within the UQSH framework, the Qu-foam remains the same field medium, while different kinds of radiation form tension fronts of different strengths within it.

The higher the frequency and energy density, the more tightly timed and more strongly tensioning the front acts; the lower the frequency, the broader, softer, and more thinned-out the same basic motion of the field medium appears.

In this picture, space is not a passive background. It is the dynamic medium in which field states absorb, transmit, and reorder tension. Radiation arises when a locally bound or excited state releases its tension into the surrounding field as a propagating front. The speed of light does not primarily describe the velocity of a transported object, but the maximum reorganization rate of the field. It is the limiting rate at which a change in tension can be transmitted from one field region to the next.

It is important to distinguish between the carrying tension front and its spectral structure. The front itself is not simply color or a spectral line. The spectrum arises as an internal modulation of this front through interaction with bound field states such as atoms, molecules, plasma, stars, or more complex structures. What we perceive as light, color, or spectrum is therefore not a complete view of the front, but the modulation that can couple to our observational regime.

At first, this idea may seem unfamiliar, but it connects directly to everyday observation. We do not see things because their complete field reality enters our eyes. We see the modulations of a radiation front that has been absorbed, altered, re-emitted, or scattered by matter. A red object is not red because it emits an isolated red substance, but because it absorbs certain modulation components and couples others back into our visual regime in a stable way. This simple observation is generalized here: every spectral observation is a coupling to a modulation layer of a carrying front.

For this purpose, the UQSH uses a scale-relative language. Our directly observable regime is only one organizational region of the field and serves as the first basis of comparison. The immediately lower connecting scale is referred to as the *subregime*, and the immediately higher one as the *superregime*. These terms do not refer to separate worlds, but to recursively effective organizational layers of the same field medium. Each regime level can act on its neighboring level and be influenced by it, but only in the coupling form readable for that regime.

A structure may therefore appear differently depending on the observational and coupling regime: in our regime, it may appear as a spectrum; in a subregime, as an independent field node; and in a superregime, as a modulation of a larger carrying tension front. Subregime and superregime thus do not designate absolute locations, but relative connecting scales of field organization: the lower and the higher regime with which a given observational regime is directly and recursively coupled.

This work brings together motifs that had already appeared separately in earlier UQSH works: light within light, field-mechanical redshift, the CMB as a coupling band, matter as bound field organization,

the dark-matter effect as field tension, black holes as saturation nodes, magnetism, chemistry, and regimes as scale-relative readability. What is new is the shared placement of these motifs within an ontological field framework, in which light, energy, gravitation, matter, time, and observation are understood as different readings of a deeper Qu-foam dynamics.

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Glossary of Terms

Qu-foam The term for the universal, continuous fundamental field of the UQSH. The term *Qu-foam* denotes the dynamic, multi-scale, foam-like organized ground structure from which radiation, matter, gravitation, tension, relaxation, and observation emerge as state forms.

The expression *field medium* or *field* is used throughout this work as an additional way of describing the physical function of the Qu-foam in more accessible terms: it is the carrying medium in which field motion, coupling, saturation, dedensification, and reorganization take place. *Field medium* is therefore not a second name for a different object, but a functional description of the Qu-foam.

What is meant is neither empty space nor a substance in the classical sense, but the dynamic ground structure from which all physical properties emerge. The concept of the Qu-foam stands, terminologically, in the tradition of Wheeler’s quantum-geometrical considerations [19], but is interpreted here as the UQSH-specific continuous ground structure.

Tension front A propagating field motion that arises from local excitation, release, or reorganization of the Qu-foam. In the main text, radiation is developed as a spherically propagating tension front.

Modulation curvature The internal imprint or fine structure of a tension front. It denotes the regime-compatible order that can later become readable as spectrum, frequency structure, or path information.

Regime A scale-relative organizational domain of the field, with its own coupling capacity, stability, saturation, and observability. A regime is not a separate world, but a specific layer of readability within the same field medium.

Subregime A lower organizational domain of the field relative to our observational regime. Structures of a subregime may not be directly visible to our regime, but they may become effective as internal modes, fluctuations, coupling thresholds, or bound fine structures.

Superregime A higher organizational domain relative to our observational regime. A structure that appears in our regime as a large-scale node may be read in a superregime as a smaller modulation, substructure, or coupling unit.

Anektron A hypothetical minimally saturated field mode of a regime. The Anektron does not denote the electron itself and does not replace the established concept of the electron. It describes the first independent coupling unit in which field motion is bound and stabilized to such a degree that it can appear as a measurable matter-capable field form.

Counter-Anektron / Antianektron The complementary counter-parity of the Anektron. It describes the positron-like counter-orientation of the same minimally saturated field-mode class. Anektron and Counter-Anektron form two coupling directions of a common minimal mode family:

$$A_n^{(+)} \quad \text{and} \quad A_n^{(-)}.$$

Anun The maximally saturated field form of a regime. The Anun denotes the upper limit state of local field saturation, where further deepening can no longer occur as an increase in local curvature. Within the UQSH framework, black holes are the most natural astrophysical candidates for Anun states.

Dedensification A field-mechanical process in which a tension front spreads over larger regions and thereby loses local tension density, timing density, and regime-compatible storage density. Observationally, dedensification may appear as weakening, redshift, or loss of fine modulation structure.

Saturation A state in which a field region can no longer carry additional field motion as further local deepening or densification. Saturation does not bring dynamics to an end; rather, it leads to reorganization, boundary formation, release, recoil, re-emission, or regime transition.

Coupling filter The regime-dependent selection of what can become readable at all for an observer or measuring system. Not every existing field organization couples to every regime. What becomes visible is only what has the coherence, frequency, density, duration, and geometry that match the respective coupling filter.

Maximum gravitation The maximum locally effective gravitational depth of a regime. It does not mean infinite gravitation and does not imply a singularity, but denotes the limit at which additional field tension no longer appears as further local depth. The depth saturates, while range, boundary structure, and release dynamics may continue to grow.

2. Radiation as a Spherically Propagating Tension Front

Within the UQSH framework, radiation is free field motion. A local excitation does not produce a small sphere that flies through space, but a spherical response of the medium itself. In the idealized case, this produces not only a single front, but a dense sequence of successive spherical tension fronts. One may picture this as a sphere within another sphere, within another sphere: each new local excitation triggers a new field response, while the previous fronts are already propagating onward at the maximum reorganization rate of the field.

The excited region reorganizes its neighborhood, this neighborhood in turn reorganizes the next surrounding region, and in this way a continuous sequence of tension fronts arises. The fronts are not solid things, but dynamic patterns that maintain themselves because the medium passes on its local order step by step. Each individual front propagates at the speed of light c . What differs is not the speed of the fronts, but their timing sequence: depending on the frequency, energy density, and tension strength of the source, the spherical fronts arise more densely or farther apart.

Since the source itself is usually not at rest, but part of a moving field organization, the successive fronts do not arise from exactly the same spatial state. Each new front carries with it the slightly

shifted and altered coupling situation of the source. As a result, the spherical fronts overlap into a complex source geometry. Radiation thus appears as a sequence of moving, nested field responses: each front propagates at c , while frequency, spectrum, and intensity arise from the temporal timing, the internal tension density, and the motion of the source.

This should not be pictured as a sphere flying through water, but rather as a disturbance in an elastic, multilayered fabric. The wave does not consist of a solid object that travels along with it. It arises because each region of the medium passes its local order on to neighboring regions.

In scale-relative terms, a tension front never encounters a completely passive medium. Every apparently local field region already contains internal motion, coupling, and reorganization. The front therefore does not alter a resting background, but changes the direction, density, and coupling form of a field motion that is already present at deeper levels.

In this view, radiation and matter are different state forms of the same field motion. Radiation is freely propagating field tension, while matter is locally stabilized field organization. The more precise transitions between free front, re-emission, and bound matter are discussed later in the sections on energy and matter formation.

The carrying tension front can take up internal modulations. These modulations form the spectrally readable fine structure of the front and are developed in more detail in the following sections on detection, quanta, and spectrum.

This view has an important consequence: a front may be visible to one regime while acting only as background tension for another. What is electromagnetically readable for us may appear in another regime as coarse gravitational tension. Conversely, a higher-regime front may thin into our regime and only then take on a visible spectral modulation here. In this way, radiation, spectrum, and gravitation are not separated, but understood as different forms of readability of the same field motion.

2.1. Local Detection and the Wave–Particle Dualism

The previous description leads directly to a new interpretation of the wave–particle dualism. If radiation is a spherically propagating tension front of the field medium, then what propagates is not point-like. Propagation belongs to the entire front. Measurement, however, has so far never occurred at the entire front, but always only at a local coupling point.

A detector, an atom, or a molecular structure therefore does not measure the complete spherical field front. It couples only to that small region of the front that reaches its own local structure and can trigger a sufficiently strong reorganization there. The observed “photon” is, in this view, not the whole light front, but the local event in which a section of the front becomes effective in a bound state of matter.

The apparent dual nature of light therefore does not arise from two different ontological properties, but from two different levels of the same process. The wave-like character belongs to propagation: the field tension spreads spatially, overlaps with itself, interferes, and carries internal modulations. The particle-like character belongs to detection: a bound field state responds only locally and threshold-dependently. What becomes visible is therefore a point, even though the underlying field organization is extended.

This explains why individual light quanta are registered as point-like, although their propagation is wave-like. The point on a screen is not the light itself, but the local response of the screen to

the incoming field tension. The extended tension front remains present as field organization, but it becomes visible only where matter enters into resonance with it and undergoes a stable change.

This becomes especially clear in the double-slit experiment. A spherical tension front reaches the two openings and continues behind them as an overlapping field motion. The front can act through both slits because it is not a small object that would have to choose one path. It is a spatially extended reorganization of the medium. Behind the slits, an interference pattern of field tension therefore forms.

The screen, however, does not register this pattern as a continuous surface, but only as individual local coupling events. Each hit is the moment in which a small region of the screen is sufficiently excited by the incoming field organization. Over time, many such hits reproduce the interference pattern. The pattern belongs to the extended field front, the individual point to local detection.

This also makes it understandable why the spectrum cannot be regarded as an independent thing. The spectrum does not carry itself. It is the internal modulation order of the carrying tension front. This modulation can be shaped by atoms, molecules, plasma, stars, or galactic field states. But it is again measured only where a local material system couples to this modulation.

In this view, the photon is not a small object that flies through space like a particle. It is the quantized local coupling response of a bound field state to an incoming tension front. Quantization therefore does not arise as a fragmentation of the front, but as the discrete response of matter to a continuously propagating field organization.

The UQSH therefore clearly separates three levels:

Propagation = spherical field-tension front,

Spectrum = internal modulation of the carrying front,

Photonic event = local, threshold-dependent coupling to matter.

The wave–particle dualism therefore appears not as a fundamental contradiction, but as an observer effect of a regime-dependent coupling process. What propagates is field-like and extended. What is measured is local and discrete. The wave belongs to the medium, the particle event to the detector.

This formulation preserves the empirical side of quantum physics: light appears quantized in measurements, energy transfers occur discretely, and interference patterns also arise in single-emission events. The ontological interpretation, however, shifts. It is not a point-like particle that interferes with itself, but an extended field-tension front that produces an interference field from which local coupling events emerge.

In the UQSH, the photon is not a flying object, but the local coupling event of a spherical field-tension front.
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2.2. *Quanta as Minimal Coupling Events*

Within the UQSH framework, quanta are not small solid objects that fly independently through space. A quantum denotes the smallest stable coupling portion in which a field motion can be absorbed, emitted, or reorganized within a regime. Quantization therefore does not arise because the field consists of isolated grains, but because bound field states allow only certain stable transitions, thresholds, and coupling forms.

Quantum = minimal stable coupling event of a regime.

The photonic event described above is a special case of such a quantum. It denotes the local, stable coupling of a propagating tension front to a bound field state.

Within the UQSH, photonic events can be distinguished according to their coupling role:

Emission quantum = local release of a bound field state,

Absorption quantum = local uptake of a matching modulation,

Scattering quantum = local deflection or re-imprinting of a front,

Re-emission quantum = newly modulated release of previously absorbed field motion.

This distinction does not mean that different kinds of photons are being introduced as new particles. It describes different coupling roles of the same photonic event type. A photon is therefore not a small object with a fixed internal content, but the regime-compatible, local, and quantized reading of a propagating field-tension front.

The energy content of a photonic event depends on the timing density of the front. The relation $E = h\nu$ is interpreted later, in the section on energy as field motion, in field-mechanical terms as a relation between frequency, tension density, and coupling strength.

Quanta are not field grains, but minimal stable coupling events.

This interpretation connects to the earlier UQSH interpretation of *Light Bubbles*, in which the particle event was described as the local trace of a spatially extended field motion [27].

2.3. *Spectrum as Internal Modulation Curvature*

Within the UQSH framework, a spectrum is not an external property of a light particle, but an internal modulation of the carrying tension front. Atoms, molecules, plasma, dust, or other bound field organizations act as resonance structures. They cannot imprint an incoming or emerging front arbitrarily, but only according to their own stable states.

Spectral lines are therefore stable resonance responses of bound field nodes. By bound field nodes, this text refers, for example, to atoms, molecules, ions, or plasma states, that is, material organizations that allow only certain stable transitions and couplings. When a tension front interacts with such structures, a characteristic modulation order is imprinted onto it.

For us, this modulation order appears as a spectral line. It shows which resonance has taken place between the propagating front and the respective material field node. The front should not be understood as a thin line or a single ray, but as a spherically growing field surface. Every point on this surface is a local reorganization state of the medium that passes its tension on to neighboring regions. The entire front therefore forms an extended, spherical coupling surface on which the internal modulation is carried along.

The term “light within light” takes up an earlier UQSH motif, according to which a spherical field response is itself the carrier of finer internal distortions and modulations [26]. It describes precisely this layering: a larger spherical tension front carries finer modulations within itself. This finer modulation does not lie outside the front, but is an ordered tension structure within the front surface itself. It is no less real than the carrying front.

It corresponds to a smaller curvature within a larger curvature, an ordered tension within the larger propagating tension. While the front surface grows outward, this internal order remains imprinted on the front as a relational structure and is carried again with every local transmission of the field state.

Over large distances, the modulation of a tension front can be partly preserved while its overall timing scale shifts. This shift is described in the following section as field-mechanical dedensification. For spectral analysis, the decisive point here is simply this: the modulation remains readable as a relative order as long as it is carried coherently enough through propagation.

A spectrum is therefore not an unchanging imprint of the source. It is the readable residual order of a propagating field-tension front that has been modulated along the way and partly thinned out. It contains traces of the source, of the field regions it has passed through, and of the observational regime that finally reads this order.

This view also changes the meaning of astronomical spectra. When a galaxy has a spectrum, we do not see its complete field organization, but the electromagnetically readable modulation layer of its stars, gas clouds, dust components, active nuclei, and propagation paths. The deeper gravitational field configuration must additionally be inferred through dynamics, rotation curves, lensing, and large-scale field structure.

2.4. *Spectrum as a Recursive Structure*

The modulation level is itself nested. A line may appear in our regime as a simple transition, while in a subregime it may be a complex organization of finer field states. Likewise, an entire galaxy may appear in a superregime as a spectral modulation of an even larger field front. Spectrum thus becomes a scale-relative concept. It denotes not only atomic lines, but more generally the readable structure of a carrier front.

This recursive view can be stated briefly:

$$\text{Carrier front}_n = \text{Modulation}_{n+1}, \quad \text{Modulation}_n = \text{Carrier front}_{n-1}.$$

What carries in one regime is itself carried in a larger regime. What modulates in one regime may be a carrier of its own in a smaller regime. This symmetry is crucial for the UQSH because it explains why the same terms can be applied to atoms, galaxies, the CMB, and large-scale structures without identifying them as the same thing.

2.5. Observation as Coupling

In the UQSH, observation is not the passive imaging of a fully visible world. It is a resonance coupling between a propagating field structure and a bound observer system. We do not see the entire tension front. We see only the modulation layer that can couple stably to our biological, atomic, and technical structures.

Visible light is therefore not a privileged universal state, but the coupling window of our material regime. Color is not an absolute property of radiation. It arises from the way our eyes and nervous system read out certain modulation ranges of the front. Other living beings, other instruments, or hypothetical observers in other regimes would read different layers of the same field reality.

Instruments do not change this basic logic either. They only extend our coupling window. Infrared, ultraviolet, X-ray, and gamma radiation are not entirely different carriers, but different tension densities and modulation ranges of the same field propagation. An X-ray telescope reads a different coupling layer than an optical telescope, but both read regime-compatible modulations of a field front.

A simple example is how we see our surroundings. We rarely see the original solar front directly. Most of the time, we see radiation that has been reflected, absorbed, scattered, and newly modulated by matter on Earth. When we look directly at the Sun, a much denser front couples in and overloads the biological system. Starlight from great distances, by contrast, reaches us thinned out enough to be read stably.

These examples show that observation is always selection. It does not select consciously, but through coupling capacity. A detector measures only what can enter into its material state. Everything else may be real and still remain invisible to us. For the UQSH, this is not a weakness of observation, but a basic condition of all observation.

It follows that horizons must not be understood only geometrically. Within the UQSH framework, a horizon is always also a coupling horizon. This consequence is developed in the following section with reference to the cosmic microwave background and the visibility limit of our observational regime.

2.6. CMB and Visibility Limit

In the standard model, the cosmic microwave background is precisely characterized by the Planck measurements [3]. The UQSH does not dispute this finding, but interprets its coupling significance differently.

In this view, the cosmic microwave background takes on a different meaning. Within the UQSH framework, it is not the end of the universe and not a fixed wall beyond which nothing exists. Rather, it describes the lower coupling limit of our observational regime: the range in which strongly dedensified tension fronts can no longer be resolved as individual sources or clearly separated spectra, but still interact statistically with our regime.

What we measure as the CMB would then be the smoothed residual order of many thinned-out

boundary dynamics from different directions, distances, and propagation paths. Their common feature does not necessarily lie in an identical origin, but in their similar coupling state upon arrival in our regime. The carrying field motions have thinned out so far that individual sources, line structures, and fine modulations no longer remain separately readable. What remains is an almost isotropic background band of weak, statistically superposed field tension.

The visibility horizon is therefore not an absolute edge of the universe, but a coupling horizon. How far we can see depends not only on distance, but also on how much modulation order a tension front has retained by the time it arrives in our regime. Dense, tightly timed fronts can remain readable as separate signals over greater distances, while already strongly dedensified tension fronts tend to pass into a smoothed background coupling.

The more detailed field-mechanical interpretation of this dedensification, including the shift from higher to lower frequency, is treated later in the section on energy, frequency strength, and reorganization. For the CMB, the decisive point here is only this: it does not mark an absolute wall, but the range in which many strongly dedensified tension fronts couple to our regime only statistically, as a background band.

An observer at another cosmic location, for example a billion light-years away, would therefore not have a CMB horizon identical in every detail. Such an observer would measure different field paths, different local structures, different degrees of dedensification, and different anisotropies. The CMB would be statistically comparable, because it describes the same general coupling range of strongly dedensified tension fronts, but its concrete fine structure would be position-dependent, path-dependent, and regime-dependent.

The CMB thus appears not as the final boundary of reality, but as the boundary of our present spectral readability. It marks the range in which separate sources and fine modulations have largely passed, for our regime, into a shared background coupling.

The denser a front is at its origin, the longer it remains readable
as an ordered signal despite dedensification.

2.7. *Densification, Re-emission, and Matter Formation*

Dedensification is only one direction of field reorganization. The opposite direction is densification. Within the UQSH framework, every bound structure has its own field tension. This is present not only in extreme objects such as neutron stars, magnetars, or black holes, but already in ordinary stars, small red stars, hot blue stars, planetary bodies, plasma fields, and atomic field nodes. The difference is not whether field tension is present, but how strongly, densely, coherently, and sustainably it is organized.

A broadly distributed or low-frequency field motion can enter such field nodes and be densified, modulated, or re-emitted there according to the local tension strength. More weakly tensioned field nodes tend to carry and release absorbed field motion in longer-wavelength ranges, while more strongly tensioned field nodes can time it more densely and transfer it into higher frequency ranges. The concrete spectral color of a re-emission therefore depends on the local tension order of the emitting node.

The observed spectral color is therefore not only an indication of temperature, but also a sign of the field-mechanical tension order of the emitting node. Which exact frequency layer becomes visible depends on how the node re-emits the field motion it has absorbed.

2.8. Redshift as Field-Mechanical Dedensification

Within the UQSH framework, the observed cosmological redshift is not disputed. Distant sources appear redshifted, and this finding remains central. The UQSH, however, shifts its interpretation: redshift is not read primarily as the metric expansion of a passive space, but as the field-mechanical dedensification of propagating tension fronts within the dynamic Qu-foam. What appears in the standard model as a stretching of the spatial scale is interpreted here as a decrease in the local organizational density of the radiation front.

Such a tension front propagates spherically and distributes its field motion over a growing surface:

$$A(r) = 4\pi r^2.$$

As a result, the local tension density decreases. The modulation carried by the front is spread more broadly, the timing density decreases, and the frequency readable for our regime shifts toward lower ranges. This field-mechanical dedensification is precisely what appears observationally as redshift.

propagating tension front $\rightarrow$ dedensification $\rightarrow$ larger timing scale $\rightarrow$ redshift.

Redshift is therefore not only a property of the source, nor exclusively a property of an expanding space. It is the integrated coupling trace of the entire propagation history. Source, travel time, traversed voids, filaments, halos, local field tension, scattering, modulation smoothing, and the observational regime all act together on the frequency order that is finally measured.

Especially in large-scale relaxed regions, such as voids, dedensification can become more pronounced, because there is less baryonic anchoring, less field saturation, and less coherent re-imprinting. In dense filaments, galaxy clusters, or strongly tensioned halo regions, by contrast, dedensification can be inhibited, superposed, or partly remodulated. Within the UQSH framework, the observed redshift is therefore not only a distance marker, but also a trace of the field state through which the front has passed.

The radiation does not disappear in this process, but is transferred into a broader, more weakly timed, and longer-wavelength coupling form for our regime.

Redshift is not disputed;
it is interpreted field-mechanically
as the dedensification of propagating tension fronts.

2.9. Dedensification as the Distribution of Frequency Strength

As described in the section on redshift, a free tension front distributes its field motion during propagation over a growing spherical surface. For the interpretation of energy, the decisive point here is this: the same field motion is distributed over larger regions of the field medium. The local tension density decreases, the timing is carried more broadly, and the frequency strength readable for our observational regime decreases.

Dedensification therefore does not mean that energy disappears. It means that the local density of field motion decreases. An originally very high-frequency tension front can thereby gradually shift its

regime-compatible readability toward lower frequency ranges. In an illustrative form, this corresponds to a sequence such as:

$$\gamma \rightarrow \text{X-ray} \rightarrow \text{ultraviolet} \rightarrow \text{visible light} \rightarrow \text{infrared} \rightarrow \text{microwave}.$$

This sequence should not be understood as a fixed path followed by every individual radiation front, but as an example of the direction of field-mechanical dedensification: high timing density is transformed over large distances into broader, more weakly timed modulation that appears longer-wavelength to our regime.

A gamma front has very high frequency strength. Its field motion is tightly timed, locally dense, and strongly reorganizing. During propagation, this density can decrease through geometric distribution, path coupling, scattering, interaction with matter nodes, passage through relaxed field regions, and internal smoothing of modulation. The front remains field motion, but its locally readable organization becomes broader. For an observational regime, this appears as weakening, redshift, or transition into lower frequency ranges.

Dedensification therefore first describes the shift of the local timing and frequency order. The question of why no energy disappears in this process, but field motion is merely reorganized, is addressed in the following paragraph.

3. Double Gravitational Effectiveness

If spectral modulation is itself a form of tension curvature, then it is not only information. It acts field-mechanically. A propagating front therefore has a double effect. First, as the carrying front, it spans a large-scale domain of action. Second, it carries internal modulation curvatures that produce their own couplings and local field responses on finer scales.

A dedensified front therefore does not become ineffective when it falls out of our direct visibility window. Its visible modulation may disappear, while its carrying field tension continues to alter tension relations within the field. Conversely, when a higher-regime front enters our regime, it may take up new atomic, molecular, or plasma-like modulations and thereby appear as an ordinary electromagnetic spectrum.

Within the UQSH framework, spectrum therefore has a double meaning. On the one hand, it is information about the readable modulation order of a front. On the other hand, every such modulation points to real field tension that cannot be fully separated from gravitational field response.

$$\text{Radiation} = \text{carrying tension front} + \text{intrinsic modulation curvature}.$$

$$\text{Observed spectrum} = \text{regime-compatible reading of this modulation curvature}.$$

$$\text{Gravitational effect} = \text{field response to carrier front and modulation curvature together}.$$

This view also explains why electromagnetic spectra alone do not have to contain the complete field configuration of a galaxy. Light shows the readable modulation layer. The gravitational field

organization must additionally be inferred through rotation curves, lensing, dynamics, baryonic structure, and the large-scale environment.

3.1. Energy as Field Motion, Frequency Strength, and Reorganization

Within the UQSH framework, energy has a simple common denominator: energy is field motion. It is not an independent substance added to the field, but the effective activity of the field itself. Wherever the field oscillates, tensions, binds, curves, carries, relaxes, or reorganizes itself, energy appears.

This definition connects phenomena that are otherwise treated separately. Light is freely propagating field motion. Matter is bound and stabilized field motion. Gravitation is organized field tension and curved field motion. Mechanical motion is the displacement or reordering of a field pattern. Temperature is the statistically readable agitation of many small field motions. Inertia is the resistance of bound field motion to forced reorganization.

The energy content of a field motion is expressed in its strength, density, and timing. In freely propagating radiation, this timing appears as frequency. The familiar relation

$$E = h\nu$$

is read field-mechanically within the UQSH framework: the frequency ν describes the timing density of the propagating tension front. The higher the frequency, the more densely and strongly the local reorganization of the field is timed; the lower the frequency, the more dedensified and weaker the field motion readable for our regime becomes.

Frequency is therefore not merely an external wave property, but a measure of the energy content of free field motion. High-frequency radiation carries more energy because it engages the field at shorter intervals and with higher reorganization density. Low-frequency radiation carries less energy because its tension front is broader, more relaxed, and less densely timed.

For bound states, the same relation appears in another form. Matter does not possess energy primarily as a freely running frequency, but as bound, circulating, and stabilized field motion. Its frequency strength is then not directly visible as light frequency, but is organized as internal timing, binding energy, mass, inertia, and coupling capacity.

Energy is therefore not any one of these phenomena alone, but their common origin. It is what underlies all state forms: the field in motion. What differs is only the degree of binding, direction, coherence, saturation, timing, and coupling form of this motion.

$$\text{Energy} = \text{field motion.}$$

$$\text{Energy content} \sim \text{strength, density, and timing of field motion.}$$

$$\text{free field motion} \rightarrow \text{frequency} \rightarrow \text{radiation energy,}$$

$$\text{bound field motion} \rightarrow \text{internal timing} \rightarrow \text{mass, inertia, and binding energy.}$$

3.2. *Energy Distribution and Reorganization*

If energy is field motion, then within the UQSH framework it cannot simply disappear. It can, however, change its form, lose its local readability, dedensify, bind, scatter, relax, or pass into other coupling regimes. A free tension front does not lose its field character in this process; it distributes its motion over larger field regions. What disappears is not the energy itself, but a particular order in which it was recognizable to an observational regime.

Conversely, distributed field motion can be refocused and bound again under suitable conditions. When it encounters tensioned field regions, matter nodes, plasma, magnetic fields, filaments, stars, accretion zones, or Anun environments, it can be densified again, modulated, or transferred into bound organization. The same field motion then no longer appears as free radiation, but as heat, acceleration, chemical bonding, matter formation, gravitation, or high-frequency re-emission.

In this view, conservation of energy therefore does not mean that a specific form of appearance remains unchanged. It means that the underlying field motion does not fall out of the field. It is reorganized between different state forms:

$$\begin{aligned} \text{free field motion} &\Rightarrow \text{bound field motion} \Rightarrow \text{tensioned field organization} \\ &\Rightarrow \text{relaxed field motion.} \end{aligned}$$

In established language, this corresponds approximately to transitions between radiation, matter or binding energy, fields, and thermal or diffuse background energy:

$$\begin{aligned} \text{radiation} &\Rightarrow \text{matter / binding energy} \Rightarrow \text{field energy} \\ &\Rightarrow \text{heat / diffuse background energy.} \end{aligned}$$

For a single observational regime, this redistribution may appear as loss. Radiation that dedensifies from gamma readability into infrared or microwave readability no longer has the same effect for certain couplings. It can no longer trigger processes that require high timing density. Field-mechanically, however, it has not been destroyed, but has passed into a different form of distribution and coupling.

Dedensification and densification can therefore be understood as two directions of the same conservation process. Dedensification distributes field motion over larger regions and lowers its local frequency strength. Densification brings distributed field motion back into stronger local organization. Neither process describes the creation or destruction of energy, but the reorganization of the same field motion.

Within this framework, entropy can be understood as a measure of the distribution and reorganizational freedom of field motion. A more strongly bound state has high local order, but less free capacity for reorganization. A dedensified state has lower local order, but a wider distribution in the field. An increase in entropy therefore does not mean the destruction of order, but the redistribution of locally bound organization into more broadly distributed field motion.

Within the UQSH framework, conservation of energy
is the conservation of field motion through reorganization.

4. Qu-Ground, Field Projection, and Motion: The Fundamental UQSH Equation

Within the UQSH framework, the starting point is not light, matter, or gravitation, but the Qu-foam itself. It is understood here as the deepest basis of all physical phenomena. In order to distinguish this fundamental level from its measurable forms of appearance, the notation Q is used for the *Qu-ground*.

Q does not denote a single field within the universe, nor a physical measurable quantity in the usual sense. According to the present understanding of this work, the Qu-ground has neither a definable zero, nor a definable unit, nor a location or body in space and time. Concepts such as space, time, scale, number, field state, and observation already presuppose physical readability. Q therefore does not denote an object within physics, but the prior basis from which physical readability first becomes possible at all.

The expression of a non-definable zero means here that Q cannot be set like a physical field value to an initial, empty, or reference state. Zero, unit, and scale already belong to the physically readable level of Φ .

Since Q itself does not appear directly as a single measurable quantity, its physically readable field reality is denoted by Φ . Thus, Φ is not a second object separated from Q , but the field projection of the Qu-ground into a specific regime of readability. In this projection, the Qu-foam appears as a field medium capable of carrying motion, tension, curvature, densification, dedensification, and organization.

The term *projection* is explicitly not to be understood in the sense of an optical image, a projector image, a virtual environment, or a mere appearance. It does not mean a simulation or a less real level. Rather, Φ denotes the real physical field form in which the prior basis Q appears for a regime as field, motion, tension, curvature, coupling, and organization.

This work does not claim a direct physical measurement link between Q and Φ . According to the present state of knowledge, not even the difference between Q as Qu-ground and Φ as physical field reality can be fully formalized according to the standards of our universe. Every measurement already presupposes space, time, coupling, scale, and an observational regime, and therefore lies on the level of Φ , not on the level of Q itself. What can initially be formalized are therefore only the field states, motions, curvatures, couplings, and reorganizations within Φ .

The first fundamental UQSH statement is therefore:

$$\boxed{Q \rightarrow \Phi} \quad (1)$$

In words:

$$\boxed{Q \text{ is the Qu-ground; } \Phi \text{ is its physically readable field reality.}} \quad (2)$$

In this view, physics does not begin with finished objects, forces, or particles. It begins where Q becomes physically readable as Φ . What we describe as field, space, light, energy, gravitation, or matter are therefore not independent fundamental substances, but states, motions, and organizational forms of this field reality.

The decisive next step is motion. Without motion, there is no dynamics, no energy, no propagation,

no curvature, no reorganization, and no observation. The motion of the field projection is denoted by $B(\Phi)$. It describes the original dynamics from which different physical readings arise for an observational regime.

This gives the fundamental dynamic form:

$$\boxed{Q \rightarrow \Phi \rightarrow B(\Phi)} \quad (3)$$

Here, $B(\Phi)$ is not an additional substance alongside the field projection, but its reality as motion. For the field projection itself, motion is not initially separated into light, energy, gravitation, matter, or time. These distinctions arise only when motion becomes bound, curved, timed, densified, dedensified, twisted, saturated, or observable.

Within the UQSH framework, the first physical readings of this motion are:

$$\boxed{B(\Phi) \Rightarrow \{L, E, \mathfrak{G}\}} \quad (4)$$

Here:

$$L = \text{light or free radiation}, \quad (5)$$

$$E = \text{energy or motion content}, \quad (6)$$

$$\mathfrak{G} = \text{gravitational field action}. \quad (7)$$

The direction of the arrow does not mean a temporal sequence in the ordinary sense. It means that light, energy, and gravitation are first readings of the same field motion. For our observational regime, they appear as different physical quantities. For the field projection itself, however, they are not separate things, but different aspects of the same motion.

Light is the freely propagating form of motion of Φ . It appears as radiation when field motion does not remain locally bound, but carries itself onward as a front. Energy is the motion content of this dynamics, that is, its timing, frequency, intensity, or capacity to do work. Gravitation is the curvature and tension effect of the same motion insofar as it becomes field-effective and measurable for a regime.

In simple form:

$$B(\Phi)^{\text{free}} \rightarrow L, \quad (8)$$

$$\nu(B(\Phi)) \rightarrow E, \quad (9)$$

$$\kappa(B(\Phi)) \rightarrow \mathfrak{G}. \quad (10)$$

Here, $B(\Phi)^{\text{free}}$ denotes free propagating field motion. It is read as light or radiation. The quantity $\nu(B(\Phi))$ denotes the timing or frequency order of the motion and becomes readable as energy content. The quantity $\kappa(B(\Phi))$ denotes the curvature or tension aspect of the motion and becomes readable as gravitational field action.

Further organizational forms arise from these first readings. As soon as field motion no longer propagates freely, but becomes bound, stabilized, directed, or transferred into recurring internal

dynamics, matter, inertia, magnetism, time measure, chemistry, and complex structure arise:

$$\boxed{\{L, E, \mathfrak{G}\} \Rightarrow \{M, T, \text{Mag}, \tau, C, S\}} \quad (11)$$

Here:

$$M = \text{matter}, \quad T = \text{inertia}, \quad \text{Mag} = \text{magnetism}, \quad (12)$$

$$\tau = \text{time measure}, \quad C = \text{chemistry}, \quad S = \text{complex structure}. \quad (13)$$

Matter arises as bound field-motion dynamics. It is not slowed-down motion of the field, but internally organized, bound motion whose free propagation has been transformed into circulation, binding, saturation, and local stability. From the outside, this bound dynamics appears as mass and persistent form.

Inertia arises from the same binding. It is the resistance of bound field motion to forced reorganization. A material field node already carries internal motion and order. Every external acceleration therefore requires a reordering of this bound dynamics. This resistance to reorganization appears as inertia.

Magnetism stands closer to the first readings than many later structures, but it is still not a first fundamental reading of motion itself. It arises only when field motion develops a directed twist and coherent near-order. Magnetism is therefore the directed twisting order of Φ , not simply motion as such:

$$\text{Mag} \rightarrow \text{directed twisting of } B(\Phi). \quad (14)$$

Time also does not belong to the first level. Motion is more fundamental than time. Time arises only where motion becomes readable as sequence, timing, change, or reorganization. Within the UQSH framework, time is therefore not an independent container in which the field takes place, but a measure of the reorganization of field states:

$$\tau \rightarrow \text{readability of the reorganization of } B(\Phi). \quad (15)$$

This gives the complete fundamental order:

$$\boxed{\begin{aligned} Q &\rightarrow \Phi \rightarrow B(\Phi) \\ &\Rightarrow \{L, E, \mathfrak{G}\} \\ &\Rightarrow \{M, T, \text{Mag}, \tau, C, S\}. \end{aligned}} \quad (16)$$

This equation is not to be understood as an ordinary causal equation in the sense of a temporal chain of emergence. It describes an ontological order: Q is the Qu-ground, Φ is its physically readable field reality, $B(\Phi)$ is its motion, and the known physical quantities are readings, bindings, or organizations of this motion.

The shortest core statement of the UQSH is therefore:

$$\boxed{\text{The Qu-ground is } Q; \text{ physics is the readability of its field reality } \Phi.} \quad (17)$$

Or, even more briefly:

$$\boxed{Q \text{ is the foundation; } \Phi \text{ is the readable field.}} \quad (18)$$

This avoids prematurely treating individual phenomena such as light, gravitation, or matter as ultimate fundamental quantities. They are not unimportant, but first or later readings of the same deeper field reality. For the observer, motion differentiates into light, energy, gravitation, matter, inertia, magnetism, time, and chemistry. For the Qu-ground itself, before any physical distinction, there is the possibility of becoming readable as field reality in motion and organization.

5. The Dark-Matter Effect as an Organized Field Response

Within the UQSH framework, the dark-matter effect is not understood as evidence for an additional non-baryonic matter component, but as the dynamic response of a continuous field medium to baryonic organization, radiation excitation, and local saturation. Visible matter acts not only as a classical gravitational source, but as an anchoring structure of the field. Radiation appears as a propagating tension front that continuously excites the field and produces local reorganization work. The observed gravitational excess arises from the baryon-guided field tension that does not fully relax.

Formally, the observed acceleration can first be described through an effective potential:

$$\mathbf{g}_{\text{obs}}(\mathbf{x}, t) = -\nabla \Phi_{\text{eff}}(\mathbf{x}, t).$$

This potential consists of the baryonic contribution and an additional field-organizational response:

$$\Phi_{\text{eff}} = \Phi_{\text{bar}} + \Phi_{\text{field}}.$$

The field contribution does not describe a new matter density, but a saturation-limited response of the medium to radiation and baryonic organization:

$$\Phi_{\text{field}}(\mathbf{x}, t) \sim \int \frac{Q_{\text{rad}}(\mathbf{x}', t - \frac{R}{c})}{4\pi R^2} S_{\phi}(\mathbf{x}, t) O_{\text{bar}}(\mathbf{x}, t) d^3x', \quad R = |\mathbf{x} - \mathbf{x}'|.$$

Here, Q_{rad} describes the local strength of the radiation-induced field excitation. The factor $1/(4\pi R^2)$ captures the spherical propagation of the tension fronts. S_{ϕ} describes the saturation limit of the field response, while O_{bar} describes the organizing effect of baryonic structures.

The observable excess in rotation curves can accordingly be written as the difference between the observed dynamics and the baryonically expected dynamics:

$$C(r) = v_{\text{obs}}^2(r) - v_{\text{bar}}^2(r).$$

This term does not directly measure invisible mass, but the additional dynamic effect of the organized field response. The dynamic amplification factor

$$D(r) = \frac{g_{\text{obs}}(r)}{g_{\text{bar}}(r)}$$

describes how strongly the observed acceleration deviates from the purely baryonic expectation.

Earlier works showed that this excess is not distributed arbitrarily, but possesses preferred regimes.

The normalized structural form

$$F(x) \propto x^p e^{-x/q}$$

describes whether the field response is organized more locally and concentrated, or diffusely across larger radii. Small values of q correspond to a localized peak response, while larger values of q correspond to an extended, diffuse field organization. The parameter q is therefore read as a dynamically activated transition scale, not as a freely arbitrary halo parameter.

The UQSH thus connects three levels: baryonic anchoring, radiation-driven field excitation, and saturation-limited reorganization. The dark-matter effect arises where these three levels act coherently together. It is not an additional substance in space, but the regime-dependently visible gravitation of a deeper field organization.

Within the UQSH framework, dark matter is not a missing substance, but organized field tension.

Baryonic matter anchors the field; radiation excites it; saturation limits the response.

A more detailed derivation of this effective field formulation was developed in [25]; the empirical regime structure of rotation curves was discussed in [24].

5.1. *Dark Energy, Redshift, and Void Relaxation*

Within the UQSH framework, dark energy must be treated in a differentiated way. Most of what is interpreted in the standard model as evidence for accelerated expansion, and therefore as dark energy, lies in the observed redshift of distant radiation sources. This redshift is not disputed within the UQSH framework, but interpreted differently: not primarily as the metric expansion of a passive space, but as the field-mechanical dedensification of propagating tension fronts.

This shifts the role of dark energy. It appears first not as an additional global force or substance, but as the interpretation of a large-scale dedensification dynamics of the Qu-foam. Over long propagation paths, a tension front loses local timing density, modulation sharpness, and regime-compatible storage density. What is read in the standard model as the stretching of space appears here as relaxation, distribution, and a change in timing of the field motion itself.

The relaxation tendency of the field is, in principle, present everywhere. Every field state carries within it the possibility of reducing tension, smoothing modulation, broadening timing, and transferring bound dynamics into freer field motion. This relaxation, however, is not equally visible everywhere. In galaxies, filaments, clusters, halo regions, or near strongly saturated field nodes, field motion is more often bound, scattered, re-emitted, focused, or newly modulated. There, dedensification is not abolished, but it is overlaid by baryonic anchoring, plasma, radiation fields, magnetization, and field saturation.

Voids remain important in this view, but they are not the actual main cause of the dark-energy interpretation. Rather, they are the regions in which the normal dedensification of propagating tension fronts is interrupted especially little. In voids, dense baryonic anchors, strong re-emission centers, compact field nodes, and pronounced saturation zones are largely absent. As a result, a tension front

can relax more freely over large distances, broaden its timing, and lose modulation density.

Dark energy and redshift are thereby connected, but not identified with one another. Redshift is the observable trace of dedensification. Within the UQSH framework, dark energy is the cosmological interpretation of this large-scale dedensification and relaxation dynamics. Voids describe the field regions in which this dynamics can proceed especially undisturbed. They contribute to normal dedensification, but they do not explain it by themselves.

Formally, this view can be written as a path function of dedensification:

$$E_{\text{path}} = E_{\text{geom}} E_{\text{relax}} E_{\text{bind}}^{-1}.$$

Here, E_{geom} describes the geometric distribution of the tension front over growing spherical surfaces. E_{relax} describes the intrinsic relaxation of the field motion during propagation. E_{bind} describes the inhibiting, binding, or re-imprinting effect of bound structures such as galaxies, filaments, halos, plasma, matter nodes, or saturation zones. Voids correspond to regions with small E_{bind} , so that normal dedensification is less interrupted there.

This also makes clear why the UQSH does not require an additional dark-energy substance. The observed redshift remains the central finding. Its cause, however, is not sought exclusively in the global expansion of a passive space, but in the field-mechanical history of the radiation itself: in its spherical propagation, its relaxation, its passage through differently bound field regions, and its gradual dedensification.

In this reading, dark energy does not describe a force acting uniformly everywhere, but an interpretation of the large-scale behavior of propagating field motion. Where the field is strongly bound, dedensification is overlaid by coupling, re-emission, and saturation. Where the field is weakly bound, especially in voids, the same dedensification can appear more freely. The difference therefore lies not in a second form of energy, but in the local coupling environment of the Qu-foam.

Within the UQSH framework, dark energy is read primarily as an interpretation of redshift through field-mechanical dedensification; voids favor this dedensification, but are not its main cause.

Dark-matter effect $\leftrightarrow$ baryonically anchored field tension,

Dark-energy effect $\leftrightarrow$ large-scale dedensification of propagating tension fronts,

Voids $\leftrightarrow$ weakly bound regions of facilitated relaxation.

5.2. Complete Relaxation as an Unreachable Limit State

Just as an absolute final saturation of the field cannot be reached, a fully relaxed Qu-foam state cannot arise in reality either. A totally relaxed field would be a state without residual tension, without internal timing, without reorganizational capacity, and without minimal coupling. Such a state would be the inverse singularity of maximum saturation: not infinite densification, but complete dynamic

dissolution. In both cases, field mechanics would come to an end.

The UQSH therefore excludes both extremes. Neither maximum compression nor complete relaxation can occur as a real final state. The field always remains within an intermediate range of tension, residual dynamics, and reorganizational capacity. Even in the largest voids, the Qu-foam is therefore not absolutely still, but remains dynamic.

In this sense, the cosmic background radiation can be read as an indication of a minimal residual timing that is still capable of coupling. It does not mark a fully relaxed zero state, but a lower coupling range of our observational regime. Even in the most strongly relaxed regions, the field therefore remains dynamic and, in principle, capable of reorganization.

Complete relaxation would be just as singular as complete saturation.

6. Saturation, Black Holes, and the End of Singularity

If gravitation is the effect of organized field curvature, it cannot grow without limit in depth. The Qu-foam has a finite carrying capacity within a given regime. When a local saturation limit is reached, additional field motion can no longer be carried as further local deepening of the same regime curvature. The field then does not respond with infinite curvature, but with reorganization.

For black holes, the local side of saturation is especially important here: the maximum effective gravitational depth of our regime is reached without requiring a mathematical singularity. Additional field motion then no longer appears as further local depth, but as greater range, boundary structure, relief, or regime connection.

In this view, black holes do not appear as singularities, but as saturated field nodes of maximum local regime curvature. In the terminology introduced later, this upper saturation form corresponds to the Anun function of a regime. Its depth does not grow infinitely; what can grow is the region in which the field is organized at this saturation limit.

Depth saturates; range can grow.

This sentence summarizes the mechanism. A small black hole and a supermassive black hole do not differ in the sense that one would be finite and the other infinite. Both locally reach the same limiting character of field saturation within our regime. The larger black hole has a larger saturation range, a larger boundary structure, and a larger coupling surface. It is not more infinite, but more extended.

This also resolves the question of why recursive field curvatures of all regimes do not lead to a mega-gravitation. Although every organized field curvature acts gravitationally, not every curvature couples fully into the same observational regime. Each level has its own saturation, its own carrying capacity, and its own coupling filters. The observed gravitation is therefore not the raw sum of all curvatures, but the saturated and regime-filtered projection of those components that become effective for our regime.

$$G_{\text{obs}}^{(n)} \neq \sum_{k=-\infty}^{+\infty} G^{(k)}$$

but approximately

$$G_{\text{obs}}^{(n)} = \mathcal{F}_n \left[\sum_k S_k \left(K^{(k)} \right) \right],$$

where $K^{(k)}$ describes the field curvature of a regime, S_k its saturation function, and $\mathcal{F}_n$ the coupling filter of the observing regime. This equation is not meant here as a finished theory, but as a structural formula: gravitation is saturated, regime-filtered field response.

Beyond this limit, the field remains dynamic. For our regime, however, this dynamics is no longer readable as additional local deepening, but as boundary structure, inertia, relief, and possible connection coupling to other regimes.

This also explains why black holes do not simply fall into each other immediately during mergers. When two Anun states approach each other, two isolated mass points are not meeting; rather, two extended regions of nearly maximum field tension are approaching. Their outer saturation zones begin to overlap even before actual contact. The space between them thereby becomes a highly loaded field region in its own right. The field cannot absorb this additional load arbitrarily as further deepening. It must reorganize the tension into a tolerable form. This produces braking, orbital motion, oscillating approach, lateral relief, radiation, and finally a shared saturation shell. In this view, the apparent long “dance” of two black holes is not merely a delay in gravitational collapse, but the field-mechanical process through which the medium distributes, smooths, and transforms the maximum load between two Anun states into a sustainable shared order.

Where depth is saturated, approach must first be made compatible as field tension.

The prolonged and wave-emitting merger process therefore fits the UQSH interpretation: additional tension is not translated without limit into depth, but into reorganization, boundary structure, and new field order.

6.1. Supermassive Black Holes as Long-Grown Anun Nodes

Within the UQSH framework, supermassive black holes do not appear as suddenly formed singular mass points, but as Anun nodes that have grown over long periods of time. Their extreme size therefore does not have to be explained by a single primordial initial state. It is more natural to understand them as having formed over long timescales through matter intake, galactic interactions, and above all through mergers with other black holes.

This interpretation becomes especially important where known physics has difficulty allowing the rapid formation of very massive black holes within short cosmological times. Within the UQSH framework, visible distance does not automatically have to be understood as an absolute horizon of origin if redshift is also read as the dedensification of propagating field motion. An older or larger universe would accordingly provide more time for generations of galaxy formation, accretion, and mergers.

Matter intake contributes to the growth of black holes and produces accretion disks, radiation, jets, and local field reorganization. The largest mass gains, however, are more likely to be expected from mergers. When two supermassive black holes couple, mass is not merely added. Two already highly

saturated Anun zones pass into a shared, larger saturation order. The result is a more extended Anun node with greater range, a larger coupling surface, and altered boundary dynamics.

Hawking radiation does not contradict this view. A black hole can in principle lose energy, but for astrophysically large black holes this process is extremely slow compared with accretion, interactions, and possible mergers. Before such a node disappears through radiation, it will with high probability take up matter again, couple, structure its environment, or merge with another saturated node.

In this sense, black holes are not absolutely unchanging objects, but functionally extraordinarily long-lived structuring agents. They collect field tension, stabilize galactic centers, drive jets, distribute energy, generate densification and relief regions, and can thereby favor new star formation, gas structures, filaments, or compact nodes. They therefore act not only as endpoints of bound matter, but as regenerative centers of an ongoing field organization.

Within the UQSH framework, supermassive black holes are therefore not an exception to the logic of the field, but one of its most long-lived manifestations: saturation, merger, relief, and renewed structure formation form a cycle in which Anun nodes not only store, but repeatedly reorder the cosmic field. They are therefore functionally immortal life-givers.

7. Saturation, Inertia, and Regime Transition: Anektron and Anun

The previous considerations lead to a more precise interpretation of the concept of saturation within the UQSH. Saturation does not simply mean a high energy density, nor does it mean an absolute standstill of the field. It describes the state in which a field region can no longer carry additional tension motion in the same open form as before. The dynamics is not destroyed, but transferred into another form of organization: into binding, inertia, stored tension, boundary structure, re-emission, or a cross-regime form of coupling.

Saturation is therefore not rest of motion, but a change in the form of coupling. As long as a tension front propagates freely within a regime, its local transmission occurs at the limiting rate of that regime. In our observational regime, this limiting rate appears as the speed of light c . If, however, the front encounters a region of very high field tension, for example near a black hole, it increasingly loses the ability to continue as a free front of the same regime. It is not c that becomes smaller; rather, the share of the dynamics that remains readable for our regime as free c -propagation decreases.

This distinction is essential. The invariance of the speed of light is not violated by this. c describes the local maximum reorganization rate of free, regime-internal tension fronts. The apparent slowing near maximum field saturation does not concern this local limiting rate, but the readability of the open propagation form. The front is increasingly transferred into a bound, inert, and saturated field state. For an external observer, this appears as time dilation, slowing, or horizon formation. Field-mechanically, it means that open dynamics is translated into inert field organization.

It is not the speed of light that slows down; free field motion is transferred into inertia.
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In this context, inertia is not an additional attribute of matter, but the expression of stored field motion in bound form. An inert state resists rapid reorganization because its internal field tension is already stabilized and bound. Inertia is therefore the coupling resistance of a saturated or partly saturated field

state against further reorganization. Mass appears in our regime as the measurable form of such bound field tension.

This view explains why a regime transition does not have to be understood as a breaking of c . A regime has its own resolution, its own coupling threshold, and its own form of time. What is readable within our regime as fast, free dynamics may be too fine, too fast, and not directly resolvable for a coarser superregime. From the perspective of such a superregime, our electron or photon dynamics would not be a stable individual structure, but an unresolved submodulation. Only when such dynamics is transferred by saturation into a more inert, coarser, and more stable field organization can it become readable for a higher regime as a coherent coupling form.

Saturation therefore acts as a scale translator. It does not slow the field in an absolute sense, but transforms an open, rapidly timed dynamics into a bound, more coarsely readable, and more inert field order. What appears to our regime as an upper limiting state may become readable for a higher regime as the first sufficiently stable coupling form. This opens the possibility that the end of one regime is at the same time the beginning of coupling for the next higher regime.

To distinguish these two limiting forms, two terms are introduced in what follows: *Anektron* and *Anun*.

An *Anektron* denotes, in the UQSH, a hypothetical minimally saturated field mode. It is not simply a new name for the Standard Model electron. The electron remains the experimentally confirmed observed state with charge, mass, and spin. The Anektron, by contrast, denotes the proposed field-mechanical basis behind such an appearance: a minimally saturated field mode that can act as the first independent regime-forming coupling unit. The existence of Anektrons is therefore a prediction of the UQSH. It does not claim that present measurements already clearly distinguish different electron classes, but that the electron appearance may rest on a deeper class of minimally saturated field modes.

An *Anun* denotes the maximally saturated field form of a regime. In the observational regime, the clearest candidate for this is the black hole. In this view, a black hole is not a singularity, but a maximally saturated field state in which additional field motion is no longer carried as further local deepening. Instead, range, boundary structure, coupling surface, and relief channels grow. The Anun is therefore not merely an endpoint, but an upper saturation anchor of the regime architecture.

Anektron = minimal saturation; Anun = maximal saturation.
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Between Anektron and Anun lies the actual inner domain of a regime. In this inner domain, material, atomic, molecular, stellar, galactic, and satellite-like field formations arise. Not every coupling-capable structure is an Anektron. Coupling capacity alone is not enough. A dwarf galaxy may have a measurable field effect and may take on a coupling role in galactic field space similar to that of an electron in atomic field space. Yet it is not necessarily a minimally saturated regime-transition mode. Rather, it is a secondary coupling-capable field formation within an already established potential space.

This distinction is crucial. The measurement regime initially does not recognize the internal object type, but the stable total field effect. Whether a measured coupling effect arises from stars, gas, a central black hole, dark field tension, or a mixed form is initially secondary for a coarse observational regime. What is measured first is the effective field organization. Only higher resolution and additional modeling allow an internal assignment. Precisely for this reason, dwarf galaxies, satellite systems, or other substructures in galactic field space may appear functionally analogous to electron states without

themselves being Anektrons.

The role of the Anun, by contrast, is more fundamental. Without an Anun, no permanently stable large-scale potential spaces arise. An Anun collects, centers, and saturates field tension so strongly that the environment is forced into ordered paths of reorganization. In the UQSH, black holes are therefore not merely endpoints of collapsed matter, but generators and stabilizers of large-scale field architecture. Their relief zones, jet channels, boundary tensions, shock fronts, density gradients, and magnetic orders can enable new field formations. In this view, galaxies arise not in spite of their central saturated field states, but through their structuring effect.

The formation of dwarf galaxies can therefore be understood as a secondary reorganization process. They arise preferentially where sufficient coupling energy, field tension, and relief dynamics are present, for example at the edges of jet channels, in filament axes, in shear surfaces, or in galactic boundary zones structured by the activity of an Anun. A dwarf galaxy is therefore neither an upper nor a lower regime boundary form, but a coupling-capable field formation that emerges from the inner domain of a regime stabilized by Anun action.

This gives a closed scale-relative chain of emergence:

Anektron → matter formation → electrons, nuclei, atoms
→ molecules → stars and complex matter
→ black holes as Anun → jet and boundary-field relief
→ galaxies and dwarf galaxies → Anun as Anektron of the superregime.

This chain does not mean that every electron is directly an Anektron, or that every dwarf galaxy is a cosmic electron. Rather, it describes a functional order. The Anektron denotes the minimally saturated regime-transition mode. The electron is the observable standard form in which such minimal coupling might appear in our regime. Dwarf galaxies are secondary, coupling-capable field formations within a galactic potential space. The Anun is the maximal saturation form that bounds a regime upward, stabilizes it, and at the same time can translate it into the next regime.

The central idea is therefore:

**The Anektron opens the coupling of a regime;
the Anun closes it and, in the superregime,
again assumes the Anektron function.**

This also makes understandable why electrons appear in the Standard Model as so extraordinarily stable, universal, and almost indestructible. They could represent the observable surface of a deeper minimally saturated field-mode class. Likewise, it becomes understandable why black holes should not be viewed only as destructive endpoints. They are maximally saturated field forms that constitute the upper stability of a regime, generate its inner architecture, and enable new coupling-capable structures through relief.

The UQSH thereby obtains a clear regime architecture:

Regime = field domain between minimal saturation and maximal saturation.

Minimal saturation appears as Anektron. Maximal saturation appears as Anun. Between the two unfolds the observable diversity of the respective scale: radiation, matter, atoms, molecules, stars, galaxies, satellites, and dwarf galaxies. The end of a regime is therefore not the end of field motion, but its translation into the first coupling-capable form of the next regime.

7.1. Naming: Anektron and Anun

The terms *Anektron* and *Anun* are introduced in this work as independent UQSH terms. They are intended to name two complementary forms of saturation within a scale-relative regime: the minimally saturated lower coupling form and the maximally saturated upper boundary form.

The shared component *An* refers, on the one hand, to Albert Einstein, formed from the first letter of *Albert* and the last letter of *Einstein*. On the other hand, *An* stands not only for the beginning of the first name and the end of the surname, but symbolically for beginning and end at the same time. This double meaning fits directly with the UQSH, because a regime transition has precisely this structure: what appears to one regime as the end of its maximal saturation can become, for the next higher superregime, the beginning of a new coupling capacity.

The term *Anektron* combines this *An* with a phonetic proximity to the electron. This is not meant to replace the established concept of the electron in the Standard Model. The physical role of the Anektron has already been defined above as the minimal saturation form of a regime. The naming is meant only to indicate that, within the UQSH framework, the observed electron appearance could point to a deeper minimally saturated coupling function.

The term *Anun* denotes the complementary upper boundary form. It also contains *An*, supplemented by the *u* for universality and UQSH. The final *n* can also be read as a mathematical regime or dimension index. The physical role of the Anun has already been defined above as the maximal saturation form of a regime. The naming connects this boundary function with the motif of closure, transition, and a new beginning function in the next higher regime.

In this naming scheme, *Anektron* and *Anun* are not arbitrary new particle names, but two complementary boundary concepts:

Anektron $\longleftrightarrow$ minimal saturation of a regime,

Anun $\longleftrightarrow$ maximal saturation of a regime.

The Anektron describes the lower entry into independent coupling capacity. The Anun describes the upper closure of maximal saturation. The regime transition connects both boundary forms: the Anun of one regime assumes, in the next higher regime, the function of an Anektron. Beginning and end are therefore not separate from one another, but two readings of the same scale-relative field organization.

The phonetic proximity of *Anun* to ancient Near Eastern terms such as *Anunna* or *Anunnaki* is not understood here as a historical derivation. It is accidental and arose independently. It may, however, be read as a symbolic resonance. In these ancient cosmic images, upper and lower world regions appear, along with a primordial carrying water or cosmic ocean and ordering powers that mediate between these regions. The UQSH does not adopt these ideas as a justification, but the visual proximity is noteworthy: here too, the concern is a carrying ground medium, superposed or nested regime domains, saturation boundaries, and transitions through which new stable structures emerge from a deeper field

order.

This linguistic proximity is not meant to mystify the term. It does, however, make it more vivid. *Anun* thus carries several layers at once: the honoring of Einstein, the motif of beginning and end, the *u* of the universal UQSH structure, the mathematical *n* as a regime index, and a distant poetic echo of ancient creation images in which world order emerges from a carrying cosmic ground.

7.2. Counter-Anelectron and Antianelectron

Within the UQSH framework, the Anelectron has two complementary coupling parities:

$$A_n^{(+)} \quad \text{and} \quad A_n^{(-)}.$$

Here, $A_n^{(+)}$ denotes the electron-like orientation of the minimally saturated field-mode class, while $A_n^{(-)}$ describes its positron-like counter-orientation. Both forms belong to the same Anelectron class. They are not separate fundamental substances, but two complementary readings of a minimally saturated field organization.

Linguistically, $A_n^{(-)}$ can be referred to as the *Counter-Anelectron* or *Antianelectron*. The term does not replace the positron. The positron remains the experimentally confirmed particle state of the Standard Model. The Counter-Anelectron, by contrast, describes the hypothetical field-mechanical counter-parity that could underlie a positron-like coupling.

Within the UQSH framework, pair formation therefore appears as the simultaneous formation of two complementary minimal coupling forms:

$$\text{free highly densified field tension} \rightarrow A_n^{(+)} + A_n^{(-)}.$$

Annihilation does not correspond to the destruction of substance, but to the return of this complementary binding order into freely propagating field tension:

$$A_n^{(+)} + A_n^{(-)} \rightarrow \text{free tension fronts}.$$

8. Scale-Relative Atom–Galaxy Analogy

Under the scale-relative working hypothesis of the UQSH, a galaxy appears as an atom-like field organization on a higher structural scale. What is meant is a functional correspondence of the organizational form: a central field node creates a potential space in which visible coupling zones, neutral stabilizing regions, and peripheral substructures together form an element-analogous order. For our observational regime, only those components become visible whose coherence, mass, duration, and coupling geometry are sufficient. The decisive factor is therefore not the mere number of existing structures, but their effective coupling capacity.

For a galactic system, an effective satellite or electron analogy can be introduced:

$$N_{\text{eff}}^{\text{sat}} = \sum_i w_i,$$

where w_i describes the coupling strength, coherence, and stability of the respective dwarf galaxy or

satellite structure. Satellites that couple as independent field nodes to the central galactic potential space contribute substantially to this effective number. Weakly bound, transient, disrupted, or only partly coherent structures enter correspondingly only with a weighting.

A galaxy such as the Milky Way therefore has a specific scale-relative element function. This does not arise from the visible total mass alone, nor from the number of known dwarf galaxies alone. It arises from the combination of central field organization, baryonic coherence, satellite binding, halo tension, and field persistence that is not directly visible. The count of known satellites therefore forms the first observable approximation of the electron-like shell structure.

The proton-analogous side of a galactic system lies in the coherent, baryonically visible high-tension regions. These include the galactic core, the bulge, dense stellar and gas regions, and stable spiral-arm segments. These regions act proton-like because they couple outwardly, carry visible structure, and help determine the central organizational strength of the system. They form the visibly structuring core and inner order of the galaxy.

The neutron-analogous side lies in neutrally stabilizing field configurations. On the galactic scale, such contributions arise from two connected components. First, there are baryonic structures that are actually present, but do not stand out as independent, strongly luminous, or clearly delimited coupling units. Second, there are tension fields that are not directly baryonically visible: halo persistence, diffuse field organization, empty regions with connecting tension, shear surfaces, filament connections, and non-coupling transition zones.

In this reading, empty regions are neutron-like whenever they take on a neutrally stabilizing role in the overall field. They then do not act as a mere absence of matter, but as buffering, connecting, or tension-distributing field components. They separate and connect open baryonic coupling zones, mediate between spiral-arm segments, hold field tension in the halo, and contribute to the overall stability of the galactic node.

This gives the structural assignment:

Electron-analogous $\leftrightarrow$ coupling-capable dwarf galaxies and satellites,

Proton-analogous $\leftrightarrow$ visible coherent baryonic high-tension regions,

Neutron-analogous $\leftrightarrow$ neutrally stabilizing field organization that is only partly or not directly visible.

The element analogy of a galaxy is therefore reconstructed from two observational levels. The first level is the already introduced effective number of coupling-capable satellites $N_{\text{eff}}^{\text{sat}}$. It corresponds to the electron-like shell structure. The second level is the inner distribution of visible baryonic coherence and neutral field persistence:

$$Z_{\text{eff}}^{\text{gal}} \sim \mathcal{K} [\rho_{\text{bar}}^{\text{coh}}, \Phi_{\text{core}}, \Phi_{\text{arms}}] ,$$

$$N_{\text{neutral}}^{\text{gal}} \sim \mathcal{K} [\rho_{\text{bar}}^{\text{non-coupl}}, K_{\text{halo}}, K_{\text{void}}, K_{\text{fil}}] .$$

Here, $Z_{\text{eff}}^{\text{gal}}$ denotes the effective visible core and structural coupling of a galaxy. $N_{\text{neutral}}^{\text{gal}}$ denotes the neutrally stabilizing field organization that couples to our observational regime either indirectly or not at all.

A galaxy thereby appears as an enlarged atom-like field organization. The reason lies not in identical

particle numbers, but in the recurring working mode of the field: central densification creates a potential space; visible coherent structures couple outwardly; neutral field components stabilize; and peripheral substructures occupy preferred coupling regions. Because the field works according to the same organizational principles across all scales, galaxies of different sizes and forms arise with recurring classes, transitions, and stability patterns.

The central question is therefore: What effective scale-relative element function does a galactic field node have if only those core, shell, and neutral components are counted that couple coherently enough for our regime?

What is counted is not everything that exists, but what is coherently capable of coupling.

8.1. *Proton–Neutron Transitions as Role Changes of Field Organization*

Within the scale-relative UQSH framework, the distinction between proton-like and neutron-like components is not a rigid classification of objects. Protonic and neutronic states describe interchangeable coupling roles of bound field nodes. Already in the microphysical domain, neutron–proton transitions show that a field node can change its internal role. What the Standard Model describes as beta decay or electron capture appears here as a selective reorganization of the internal tension distribution:

$$n \rightarrow p + e^- + \bar{\nu}_e,$$

$$p + e^- \rightarrow n + \nu_e.$$

Within the UQSH framework, free beta decay means that a previously neutron-like, neutrally stabilized field order breaks open and develops a more open coupling asymmetry. The remaining state becomes more proton-like: it couples more strongly outward, becomes more directed, and appears more clearly as an independent structure to an observational regime. The emitted electron and antineutrino component is read as an exported counter-coupling through which the new open order is stabilized:

neutrally stabilizing field order $\rightarrow$ openly coupling field node + exported counter-coupling.

The reverse direction, electron capture, describes a neutralization. A proton-like state absorbs a suitable compensating counter-coupling. Its open coupling asymmetry is thereby balanced, and the state passes into a more neutron-like field order that is more inwardly closed and neutrally stabilizing:

openly coupling field node + compensating counter-coupling
 $\rightarrow$ neutrally stabilizing field order.

In this reading, the weak interaction describes the selective channel through which such boundary-stable field configurations can change their coupling role.

Scale-relatively, the same role logic applies on the galactic level. A baryonic structure appears proton-like as long as it is coherent enough to act visibly, luminously, and dynamically as its own coupling unit. If it becomes more diffuse, loses coherence, or has its open coupling compensated by environmental tension, it takes on a neutron-like function: it stabilizes, buffers, and connects without standing out strongly as an independent visible structure.

Conversely, an initially neutral field organization becomes proton-like when it builds up sufficient coherence, density, and coupling asymmetry. A diffuse tension region, a halo component, a filament zone, a gas-rich transition region, or a non-coupling part of a spiral arm can, under suitable conditions, tip into a visible baryonic structure: into star formation, gas densification, a satellite node, or a young dwarf galaxy. A previously neutron-like region becomes proton-like when it gains coherence, density, and visible coupling.

One example is a dwarf galaxy or a gas-rich subnode entering a spiral arm or a galactic density wave. If it encounters a previously weakly coupling-capable region there, it increases that region's density, coherence, and degree of organization. The region becomes more visible to our regime and acts more proton-like. Conversely, a formerly visible, proton-like region can recede into a neutron-like stabilizing role through dispersal, relaxation, gas loss, or compensation of its open coupling.

Proton and neutron analogy are therefore two roles within the same field mechanics:

proton-like = openly coupling, visibly structuring,

neutron-like = neutrally stabilizing, buffering, connecting.

The transition between the two roles corresponds to a change in coupling asymmetry:

neutron-like field order $\rightleftharpoons$ proton-like field order + free coupling modes.

On the galactic scale, this is a slow reorganization of visibility, coherence, and field tension. Regions that appear as dark, neutral, or not directly coupling-capable field components pass into visible structures under relief or densification conditions. Visible baryonic structures recede into a neutron-like stabilizing role through dispersal, relaxation, or coupling balance.

Proton-like and neutron-like denote interchangeable coupling roles: open visible structure on one side, neutral stabilization and buffering on the other.

8.2. *Quarks as Internal Intermediate Modes*

Within the scale-relative UQSH framework, quarks are not understood as an isolated fundamental substance, but as internal intermediate modes of strongly saturated field nodes. A proton-like or neutron-like state therefore does not consist of small solid particles, but of a closed organization of several internal tension directions.

These intermediate modes do not possess an independent stable external coupling. They become sustainable only within the complete field node. This is why quarks appear confined in the standard picture: not because they are held in place by an external force as if in a cage, but because a single internal tension direction does not form its own stable field form without the rest of the overall organization.

Protons and neutrons are therefore complete nuclear field nodes. Quarks are their internal orientation, tension, and coupling modes. The proton-like or neutron-like role arises only from the overall balance

of these intermediate modes:

internal intermediate modes → closed nuclear field organization
→ proton-like or neutron-like external role.

Scale-relatively, this can be transferred to galactic structures. Within a spiral arm, a bulge, or a dense star-forming region, local vortices, shear flows, gas densifications, magnetic-field directions, and not directly visible tension components can be read as quark-like intermediate modes. On their own, they do not yet form complete proton-like or neutron-like field nodes; only their shared organization produces a visible, coherent coupling form.

Quarks, or quark-like modes, therefore form the internal transition level between diffuse field tension and complete node formation:

diffuse field tension → quark-like intermediate modes → proton-/neutron-like field nodes.

9. Maximum Gravitation, Temperature, and Quasar Activity

On this basis, the question of temperature and quasar activity can be interpreted more precisely. An Anun denotes the maximum locally effective gravitational depth of a regime, but not a final standstill of field motion. When additional field tension can no longer be carried as further local deepening, it is transferred into internal tension, inertia, boundary structure, relief, recoil, and regime coupling.

The saturation of an Anun is therefore a progressive boundary process. The more strongly the field is locally loaded, the more its tendency grows not to deepen additional tension further, but to redistribute it toward the boundary, relief axes, and the surrounding field. The Anun thereby does not become a singularity, but an active center of redistribution.

This view connects directly with the interpretation of black-hole temperature. The effective temperature of an inactive black hole can be understood as a measure of the remaining open internal mobility of the field. The more strongly an Anun is saturated, the more free dynamics is bound into internal tension organization. The saturated core therefore appears not hotter, but colder: not because energy is absent, but because open oscillation, free reorganization, and thermally readable mobility are increasingly inhibited. This interpretation does not replace the standard basis of black-hole thermal radiation, which goes back to Hawking's calculation of particle creation at black holes [4].

As the size of an Anun grows, it is therefore not an infinite gravitational depth that grows, but the range of the saturated tension region. The tension is distributed over larger boundary zones, coupling surfaces, and transition regions. Maximum gravitation thereby does not become harder in the sense of a singularity, but more extended and field-mechanically softer: the field does not fully yield to the pressure, but translates it into extended saturation zones, relief channels, and reorganizational structures.

The largest Anun therefore show especially pronounced boundary and recoil effects: what appears at the center as coldness, inertia, and blocked reorganization can become visible outside as extreme activity. In this view, quasars can be understood as Anun systems operating close to the maximum load-bearing limit of their regime. The inner saturated core remains extremely cold in the sense of low open reorganizational capacity, while the surrounding region appears highly active. The observed heat

and radiation do not belong to the internal temperature state of the Anun itself, but to the outer relief and re-emission zone: accretion disks, jets, shock fronts, and magnetized plasmas are the regions in which bound or focused field tension is transferred back into propagating radiation and new matter organization.

A quasar can thereby remain long-lived. It is stabilized by the ongoing coupling between the maximally saturated Anun, the surrounding field, and relief structures. The system partly maintains itself because it relieves field tension, produces re-emission, and at the same time prevents complete relaxation of the saturated region. The possible matter-forming consequences of such relief zones are treated in the section *Matter Formation*.

In this sense, the largest black holes are not simply the “most extreme singularities”, but the largest extended zones of maximum gravitation within our regime. Their internal dynamics is strongly bound and cold, while their boundary regions are among the brightest and most productive reorganization zones in the universe.

Maximum gravitation does not stop dynamics; it binds it internally and forces relief at the boundary.
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10. Uncertainty as Room for Stability and Growth

Within the UQSH framework, uncertainty has not only epistemological, but ontological significance. It does not describe a mere limitation of our measurement, but a necessary degree of freedom of the field. A completely sharp, fully determined field state would no longer have any internal freedom. It could not yield, relax, couple, or form a new order. It is precisely the remaining uncertainty that prevents a rigid final state.

This view connects uncertainty directly with stability. A stable field organization is not stable because it is absolutely fixed. It is stable because it remains open within certain limits. It has enough binding to carry form, but also enough residual motion to redistribute tension, absorb disturbances, and enter into couplings. Stability therefore does not arise from total determinacy, but from controlled openness.

Uncertainty is thereby connected with the two unreachable limiting states of the UQSH: complete saturation and complete relaxation. Both extremes would be singular, because they would remove every possibility of further reorganization from the field. The real working range of the field lies between them: in limited order, limited uncertainty, and ongoing reorganizational capacity.

Uncertainty is therefore the condition that allows growth to remain possible. Within the UQSH framework, growth does not mean mere increase in size, but the capacity of a field state to absorb new tensions, expand its boundary structure, form couplings, and integrate into larger organizations. Without uncertainty, there would be no boundary formation, no transition regions, no regime connections, and no formation of new stability.

This applies especially to Anektron and Anun, but in a regime-dependent way. In the Anektron, uncertainty appears as the necessary openness of the minimal coupling form: it can act stably only if it is not completely point-like, closed, and absolutely determined. Its residual openness enables internal dynamics, coupling capacity, and interaction.

In the Anun, by contrast, uncertainty does not appear to our observational regime as easy mobility,

but as the boundary openness of maximum saturation. For us, an Anun appears extremely inert, time-dilated, and almost maximally bound. What remains open is not its local depth, but its boundary structure, range, relief channels, and capacity to connect to a higher regime.

From the perspective of a superregime, the same slowed tension motion may become readable differently. What appears in our regime as an extremely stretched Anun may there be temporally contracted and appear as a new minimal, still only uncertainly resolved coupling unit. In this way, the Anun of one regime can assume the Anektron function in the superregime.

Uncertainty therefore acts at both boundaries of the regime. At the lower boundary, it protects the first stable coupling form from complete fixation. At the upper boundary, it prevents singularity and enables redistribution into the next higher regime. Anektron and Anun are therefore not only saturation boundaries, but also uncertainty boundaries: they show that stable field organization always requires a residual openness.

The formation of the world as a whole also rests on this principle. Matter, chemistry, life, and cosmic structure can arise only because field states neither dissolve completely into free flow nor become completely rigid. They remain stable enough to carry form, and open enough to allow exchange, self-reorganization, and adaptation.

The world is therefore not stable because everything is perfectly determined. It is stable because the field possesses controlled uncertainty: enough order to carry form, and enough openness to make change, coupling, and future development possible.

Uncertainty is not a flaw in order, but its room for stability and growth.

11. Space as Relative Field Order

General relativity successfully describes gravitation as the geometry of spacetime [18]; the UQSH reads this geometry as an expression of carried field tension.

Here, space is not an empty container in which things are located. Space is the relational order of the field medium itself. It describes how field states are positioned with respect to one another, how tensions are distributed, how couplings become possible, and which paths propagating field motion can take. Space is therefore not independent of the field, but the geometric reading of its current organization.

A completely empty space is not assumed in this view. Where field exists, there is tension, residual dynamics, coupling capacity, and the possibility of reorganization. Even apparently empty regions such as voids are not nothing, but weakly bound, more strongly relaxed field orders. They have less baryonic anchoring and lower saturation, but still possess a field structure through which tension fronts can propagate, dedensify, and be modulated.

Space is therefore relative to the respective regime. A structure that appears in our regime as extended space between galaxies may be readable in a superregime as an internal modulation zone of a larger field front. Conversely, a field structure that is microscopic for us may represent its own spatial organizational domain in a subregime. Space is thus not an absolute stage, but the scale-relative geometry of field organization.

Space = relational order of coexisting field states.

The spatial extension of a structure therefore describes not only its geometric size, but also its coupling range. A field node possesses space by distributing tension, creating bindings, forming boundaries, and including other field states in its organizational form. Space is therefore the visible order of these coupling and tension relations.

Within the UQSH framework, space is therefore not absolute, but field-relative. It is the way in which the field orders its states with respect to one another and makes them geometrically readable for a regime.

Space is not the container of the field, but the order of its couplings.
--

12. Speed of Light, Scale-Relative Timing, and Time Dilation

Within the UQSH framework, the speed of light and time are not quantities of the same kind. The speed of light c denotes the maximum reorganization rate of free field motion. It is the limiting motion of the field itself: the most open form in which the field can pass its local order on to neighboring field regions.

c = maximum reorganization rate of free field motion.

Time must be distinguished from this. It is not an independent background quantity and not a second constant alongside c , but the emergent measure of the sequence of field reorganizations. Wherever field states change, couple, relax, bind, or dedensify, a temporal order arises.

Time = order of field reorganization.

Without field change, there would be no physically readable time.

Thus, c is more fundamental than the respective time form of a regime. Time can be stretched, slowed, or read scale-relatively; the limiting rate of free field motion is preserved.

This distinction is especially important for scale relativity. Each regime has its own internal timing. A process that appears extremely stretched in our regime may proceed normally in its own regime. Conversely, a process that takes a long time for a deeper regime may appear almost instantaneous to a higher regime. This does not mean that the speed of light differs between regimes. It means that the temporal readability of field reorganization is scale-relative.

scale-relative time $\neq$ variable speed of light.

Rather, scale-relative time means different timing of bound field organization.

A local observer always measures light at the same speed, because their matter, their clock, their

measuring rods, and the local tension front belong to the same field regime. For that observer,

$$c = \text{constant.}$$

An external observer, however, may see that the time form of another regime is extremely stretched. The processes there appear slower, not because c has become smaller, but because the bound field organization performs fewer free reorganization steps per external comparison.

In strongly tensioned or saturated field regions, this difference becomes especially clear. The more strongly field motion is bound, the less freely it can reorganize. A larger share of the motion is held in tension, curvature, inertia, or saturation. For external observers, the passage of time there appears slowed. For the affected local observer, however, their own timing remains consistent.

stronger field binding $\rightarrow$ stretched local timing $\rightarrow$ time dilation.

Time dilation therefore does not arise from a local reduction in the speed of light. It arises from the different reorganizational order of bound field states. Light remains the limiting motion of the field. Time is the way bound systems count this motion within their own internal order.

At the Anun, this difference reaches its strongest form. Field motion is maximally bound and appears to external observers extremely slowed or almost frozen. Nevertheless, c is not suspended. The limiting dynamics of the field remains, but it no longer appears as freely outward-continuable propagation; instead, it appears as internally bound, inert, and boundary-uncertain field organization.

Anun $\rightarrow$ maximum field binding $\rightarrow$ extreme timing dilation $\rightarrow$ no violation of c .

This gives a clear ordering of the concepts. Space describes the coupling order of the field. Time describes the reorganization order of the field. The speed of light describes the limiting rate of free field motion. Space and time are therefore regime-dependent readings of the field, while c is the constant limiting dynamics by which these readings first become measurable at all.

Space = coupling order of the field,

Time = reorganization order of the field,

c = limiting motion of the field itself.

The constancy of the speed of light and the relativity of time therefore do not contradict one another. They belong together. Precisely because c is the constant limiting rate of free field motion, different regimes can time their bound reorganization differently relative to it. Time thereby appears scale-relative, but not as an independent fundamental substance. It is an emergent measure arising from field motion itself.

The speed of light is the limiting motion of the field; time is the emergent measure of its reorganization.
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12.1. Spacetime as Shared Field Readability

Space and time are therefore not discarded, but interpreted field-mechanically. Space describes the coupling order of coexisting field states, while time describes their sequence of reorganization. What is geometrically unified as spacetime in relativity appears in the UQSH as the dynamic readability of the field medium: field states have spatial extension because they distribute couplings and tensions; they have temporal duration because they reorganize their order step by step.

13. Entanglement and the Continuity of the Field Medium

From the UQSH perspective, the so-called spooky action at a distance of entangled states suggests that space should not be understood as an absolute stage. When two entangled states show shared correlations across large spatial distances, this correlation does not have to be interpreted as signal motion through space. It may instead indicate that both states remain jointly anchored in a deeper field organization.

In the standard picture, entanglement does not allow the usable transmission of classical information faster than light. This restriction is retained within the UQSH framework. The interpretation, however, shifts: no signal moves instantly from one place to another; rather, spatially separated measurement events appear as local readings of a shared deeper field order. Distance belongs to the space-time reading of our regime, while the underlying field relation does not necessarily have to be organized in the same spatial form.

Space thereby becomes relative to coupling. For our observational regime, the field appears spatially extended, temporally ordered, and causally limited by the speed of light. At a deeper level of the Qu-foam field medium, however, the same organization may act as pre-spatial or less spatial: not in the sense that it violates space, but in the sense that it first brings forth space as a regime-dependent form of appearance.

Within the UQSH framework, it follows that no point in space and time exists in complete isolation. What appears to our observational regime as spatially separated, temporally delayed, or connected only along light paths remains, at the level of the underlying field medium, part of a single continuous organization. This connection does not mean that classical information can be transmitted arbitrarily fast between arbitrary points. It means that all points are related, mediated, and not absolutely separated in their possibility of coupling, because they are states of the same field.

The apparent separation of places arises through the space-time reading of our regime. The deeper field order, however, is continuous. Every place is therefore both a local expression and part of the whole: locally limited in its observable coupling, but not ontologically cut off from the rest of the field.

Every point is locally distinguishable, but not separated from the field whole.

14. Galaxies as Spectral Field Nodes

The same logic can be transferred to galaxies. In the usual picture, galaxies are systems of stars, gas, dust, plasma, magnetic fields, and a dark-matter halo. Within the UQSH framework, they additionally

appear as large-scale field nodes with their own tension organization. Their electromagnetic spectra show only one readable layer of this organization.

Stellar populations, gas clouds, dust, star-forming regions, central nuclei, supernovae, pulsars, magnetars, jets, and active black holes jointly modulate the total radiation and the gravitational field tension. The galaxy as a whole can therefore be regarded as a spectrally modulating field node.

The analogy to atoms is structural. Atoms have characteristic spectral lines because their internal bound states allow only certain resonances. In the superregime, galaxies could analogously have characteristic organizational signatures that do not consist of individual atomic lines, but of the coarse form of their spectral and gravitational field organization.

The following table summarizes this analogy as a preliminary UQSH classification. It orders galactic systems not by new particle types, but by their broad modulation, reprocessing, and activity character.

This classification is not meant to claim a chemical periodic table of galaxies. It serves as an analogy for an ordering principle: stable spectral signature, internal organization, coupling capacity, and reaction behavior can be comparable across scales without physically identifying the objects with one another.

UQSH class	Galactic field node	Modulation character
G-I	Elliptical galaxies / bulges	smooth, old, broadly organized spectra; stabilized ground-state nodes
G-II	S0 / early spirals	medium modulation; transitional structure
G-III	Sb–Sc spirals	high local fine structure; active modulation nodes
G-IV	Starburst galaxies	strong line and reprocessing imprint; excited nodes
G-V	dust-rich IR galaxies	global spectral redistribution; obscured reprocessing
G-VI	AGN / Seyfert	core-dominated high-energy imprint
G-VII	Quasars	extreme central activity; overcritical field nodes
G-VIII	Supernova-dominated systems	pulse-like tension discharge
G-IX	Merging galaxies	unstable, strongly superposed reorganization

Table 1: Preliminary UQSH classification of galactic field nodes as a scale-relative analogy to spectral organizational classes.

14.1. Which Observational Levels Are Relevant for the UQSH

In this work, the UQSH does not claim to measure the complete field configuration of a galaxy directly. According to the present state of knowledge, such a configuration is not immediately observable. What is accessible are only quantities that appear within our coupling regime as dynamics, light distribution, spectrum, or structure. In what follows, these quantities are not understood as the complete field organization itself, but as readable traces of a deeper field order.

These include, in particular, rotation curves, baryonic structural quantities, gas distributions, photometric colors, UV and IR proxies, lensing signals, and galaxy spectral templates. Each of these observational levels shows only one aspect: rotation curves capture the dynamic effect, photometry and spectrum capture the electromagnetically readable modulation, gas and dust data show coupling

and reorganization regions, and lensing points to the integrated field curvature along the line of sight. The data used here were not generated as new observational data, but derived from publicly available astronomical datasets and catalogs. For the rotation-curve analysis, SPARC-based quantities were used in particular; additional photometric and spectral comparisons were carried out using available WISE, GALEX, SDSS, and template data. The analysis consisted essentially in relating already measured quantities to one another and testing whether, in the sense of the UQSH, they can be understood as different readings of a shared field organization.

It is therefore important to distinguish clearly: the electromagnetic spectrum is not itself the proposed superregime spectrum. It is the observable modulation layer of a galactic field node in our regime. The dynamic field response, by contrast, becomes visible indirectly through rotation curves, D_{DM} , lensing, or other gravitational observables. Only the combination of both levels approaches the actual field organization.

14.2. SPARC, q , and Baryonic Organization

One important test concerns the dynamic switching parameter q , which was obtained from rotation-curve analyses. In earlier UQSH works, q was interpreted as a dynamically activated transition scale. The central question was whether q is merely a fitting artifact, or whether this parameter is connected to real properties of galaxies.

The comparison with SPARC properties shows that q is not simply determined by morphology, distance, or disk inclination. Instead, systematic differences appear in structural quantities such as disk surface brightness, baryonic compactness, gas-fraction proxies, $3.6\text{-}\mu\text{m}$ luminosity, and D_{DM} . This difference becomes especially clear between the peak regime and the diffuse regime.

The peak regime shows, in the median, a higher dynamic excess and higher gas-fraction proxies. The diffuse q -regime, by contrast, is baryonically more compact, more luminous, and poorer in gas. The conceptual distinction is important: *diffuse* here does not describe the visible baryonic distribution, but the form of the additional dynamic field response. A baryonically compact galaxy can therefore have a diffuse q -response.

Within the UQSH framework, this suggests that the dark-matter effect depends not only on total mass, but on the way in which baryonic matter organizes the field medium. q then does not describe a simple morphological type, but a dynamic form of reorganization.

The rotation-curve basis used here follows the SPARC database [1]; the comparison with the baryonically expected dynamics is also closely related to the Radial Acceleration Relation [2].

14.3. WISE, GALEX, and SDSS

The tests with WISE, GALEX, and SDSS proxies show a different picture. Simple broadband colors do not robustly separate the q -regimes. WISE colors such as $W1 - W2$, $W2 - W3$, and $W3 - W4$ show no strong correlation with q . GALEX NUV likewise shows no significant connection. The available SDSS colors are also insufficient to clearly distinguish the dynamic regimes.

This is not a failure, but an important limitation. q is evidently not a simple color, dust, or star-formation indicator. The dynamic field response is more strongly connected to spatial baryonic organization than to global broadband colors. A deeper spectral test would require integrated spectra, line ratios, D_{4000} ,

star-formation rates, metallicities, and spatially resolved spectral data.

14.4. Multiscale Galaxy Spectral Templates

The template tests were considerably more interesting. Here, not only individual colors were considered, but continuous galaxy spectra. The spectra were brought onto a common wavelength range, normalized, and smoothed on several scales. This made it possible to test whether the spectral identity of a galaxy depends only on narrow lines, or whether a coarse large-scale spectral shape is also preserved.

The central result is this: the template classes remain distinguishable even after strong smoothing. Elliptical and S0 templates remain red-weighted and smooth. Spiral templates form an intermediate class. Starburst templates remain bluer, more activated, and more strongly modulated systems. The sequence

$$\text{elliptical/S0} \rightarrow \text{spiral} \rightarrow \text{starburst}$$

remains visible even when a large part of the fine structure is suppressed.

For the UQSH, this finding is important. A possible superregime spectrum of a galaxy would not have to resolve the atomic lines of our regime. It could instead read the broad field-tension organization of the entire galactic node. The smoothing analysis shows that such a coarse organizational form is indeed preserved in the electromagnetic templates.

The data basis of this template test was small and is not sufficient for a definitive galaxy classification. Nevertheless, the test is methodologically useful. It shows that galactic spectral identity does not lie only in narrow lines, but also in robust large-scale forms. Precisely such forms could serve as proxies for a scale-relative field-node signature.

15. Matter Formation

Within the UQSH framework, matter formation is not tied exclusively to maximally saturated Anun states. In principle, it can occur in any sufficiently tensioned field region: in red dwarfs, ordinary stars, hot stars, accretion zones, neutron stars, magnetars, jets, quasar environments, and finally in the relief fields of an Anun. What matters is not only the absolute size of the field node, but whether field-saturating radiation selectively meets other radiation modes or bound field structures. When phase, energy, geometry, orientation, and local carrying capacity fit together, free field tension can pass into the first bound coupling forms.

The most elementary expression of this transition is the formation of electron–positron pairs. Within the UQSH framework, this corresponds to the first stable splitting of free, highly densified field tension into matter-capable counter-modes. The difference between ordinarily tensioned fields and maximally saturated Anun environments therefore lies not in the principle itself, but in the intensity and frequency of the required conditions. In less extreme stellar or plasma fields, such processes may occur rarely, locally, or only for short durations. In Anun, quasar, or jet environments, by contrast, the radiation densities, frequencies, tension gradients, and coupling geometries are so extreme that the probability of pair formation and subsequent matter organization is greatly increased.

Anun states are therefore not the only places of matter formation, but the strongest natural amplifiers

of this process. Their maximum field saturation creates relief zones in which high-frequency radiation, strong magnetization, plasma, shock fronts, and dense coupling fields act together. Quasar environments are therefore especially suitable regions in which pair formation and early matter organization can occur not in a fundamentally different way, but more frequently, more densely, and across larger spatial and temporal domains.

Observationally, this is compatible with the frequent high-energy processes, ionized gas regions, hydrogen and helium lines, and indications of pair plasmas in quasar and jet environments. Within the UQSH framework, these findings are not understood as direct proof of new matter generation, but as compatible indications that highly tensioned relief fields may favor minimal matter-capable modes.

15.1. Dwarf Galaxies as Secondary Matter and Field Nodes

A natural consequence of this view concerns the origin of young, gas-rich dwarf galaxies in the vicinity of larger galaxies. If matter formation is possible in strongly tensioned relief fields, then dwarf galaxies do not have to be understood exclusively as originally small galaxies or as old remnants of early structure formation. They may emerge as secondary field organizations from the relief zones of larger galactic nodes.

Within the UQSH framework, central black holes, active galactic nuclei, jets, accretion zones, and galactic tension axes generate large-scale regions of increased field tension. In such regions, free or strongly densified radiation can selectively couple with other radiation modes, plasma, magnetic fields, and existing field structures. Under suitable conditions, such relief zones can favor minimal matter-capable coupling forms and early gas structures. If these are further stabilized, gas clouds, star formation, and later dwarf galaxies can emerge from them.

Hydrogen- or helium-rich dwarf galaxies near large galaxies therefore do not necessarily appear as a puzzle. They could be local products of an ongoing field-mechanical relief process. The main galaxy does not act only through gravitational attraction, but also structurally: its central and peripheral tension fields create the conditions under which new small field nodes arise, bind gas, and gradually develop their own coupling capacity.

Within the UQSH framework, such a dwarf galaxy would therefore not be a mere clump of matter, but a newly stabilized galactic field node. Its high fraction of hydrogen and helium would indicate that it is in an early stage of matter organization. Its proximity to the main galaxy would not be accidental, but an indication that such objects may have emerged from its large-scale tension, jet, or relief environment.

This also connects to the analogy with electron formation on a smaller scale. Just as an electron-like mode becomes stable only when free field motion passes into a bound coupling form, a dwarf galaxy can likewise arise as a larger, scale-relative coupling unit. It forms where free or diffuse field tension is transferred into long-lived organization through suitable geometry, density, and duration.

15.2. Quantum Numbers as Stable Field Markers

Within the UQSH framework, quantum numbers are not understood as externally attached properties of isolated particles, but as stable markers of bound field modes. They describe how a field organization carries mass, couples outwardly, is internally oriented, and can occupy only certain stable states. In this view, quantization means that the field cannot permanently carry every arbitrary intermediate

form, but only those organizational states whose tension, geometry, phase, and coupling fit together in a stable way.

Mass appears here as the bound field tension of a stabilized node. It is a particularly stable form of regime-effective field curvature, but not the only source of gravitational field response. A mass-carrying state is a field motion that does not propagate freely, but is locally held, organized in circulation, and stabilized against external reorganization:

$$\text{Mass} = \text{bound field tension.}$$

Charge describes the directed coupling asymmetry of such a field node. A charged state does not couple neutrally in all directions, but possesses a sign, orientation, or counter-coupling structure through which it responds selectively to electromagnetic field organizations:

$$\text{Charge} = \text{coupling asymmetry.}$$

Spin is not understood as the classical rotation of a small body, but as the internal orientation, chirality, or circulation order of a bound field mode. It describes how the internal field motion is nested in such a way that it allows only certain stable orientations and coupling responses:

$$\text{Spin} = \text{internal orientation or circulation order.}$$

Further quantum numbers can accordingly be read as markers of stable coupling roles. They indicate which internal tension directions, parities, symmetries, binding modes, or regime connections a field state possesses. Within the UQSH framework, quantum numbers are therefore not abstract labels, but readable signatures showing that a field node can exist only in certain stable organizational classes.

Quantum numbers are stable markers of bound field organization.

16. Chemistry as Flexible Field Coupling

Within the UQSH framework, chemistry describes not only the interaction of finished particles, but the stable coupling of bound field nodes within a middle range of saturation. Atomic nuclei represent highly densified, strongly bound field organizations. Electrons, or electron-like coupling forms, by contrast form the more flexible outer coupling layer through which atoms can enter into relation with one another. Only this outer, electromagnetically ordered shell makes it possible for individual atoms to form molecules, materials, and more complex structures.

The electromagnetic interaction occupies a special intermediate position here. It is not as locally locked as the strong interaction in the nucleus, and not as freely large-scale as gravitation. It forms a range of medium saturation: bound enough to carry stable forms, but open enough to allow coupling, exchange, polarization, reaction, and reorganization. This is precisely the intermediate range in which the diversity of chemistry arises.

$$\text{nuclear densification} \rightarrow \text{electromagnetic shell} \rightarrow \text{chemical coupling capacity.}$$

Chemical bonds therefore appear as stabilized field bridges between several bound nodes. A molecule is not merely a collection of individual atoms, but a shared field organization in which charge distribution, electron density, geometry, polarity, bond angles, and vibrational modes come together into a new stable overall form. The chemical identity of a substance therefore does not lie only in the atoms involved, but in the way their field shells are jointly organized.

chemical bond = stable coupling of atomic field shells.

A chemical reaction is accordingly not the creation of new substance from nothing. It is a reorganization of existing field bindings. Certain couplings are released, transition states arise, energy is absorbed or released, and new more stable or differently adapted field configurations form. What appears in chemistry as a reaction pathway is, within the UQSH framework, the path along which a system passes from one field organization into another.

old binding order → transition state → new binding order.

The energy absorption or release observed in this process describes the difference between the organizational states involved. A system can absorb field motion in order to release an old binding or reach an unstable intermediate form. It can release field motion when a new configuration is more stable and requires less free reorganization. In established language, this appears as binding energy, heat, radiation, motion, or internal vibration.

Molecular vibration thereby also receives a field-mechanical meaning. Vibrational and rotational states are not merely mechanical motions of small bodies, but periodic modes of reorganization of the entire molecular field. They show how the coupled field structure distributes, stores, and releases its tension. Spectral lines of molecules are therefore readable modulations of this internal coupling order.

Water has a special role in this context. It combines high stability with pronounced polarity, flexible hydrogen-bond formation, high heat capacity, and versatile solvent function. Within the UQSH framework, this means that water forms an especially broad coupling window for chemical reorganization. It can absorb, distribute, buffer, and mediate field motion between different molecular structures. Water thereby becomes a preferred medium in which more complex chemical order can arise and remain stable.

Chemistry is therefore the bridge between matter formation and biological organization. Matter formation produces the first stable atomic coupling forms; chemistry connects these forms into flexible reaction networks. The conditions under which such networks can pass into biological organization are discussed later in the section on the recurring working mode of the field.

matter → chemical coupling → reaction networks → biological organization.

Scale-relatively, chemistry corresponds to the general principle that bound field nodes can form new higher-order coupling forms. In the atomic regime, this occurs as molecule formation. In the galactic regime, several field nodes can form larger systems, filament connections, or coupled galaxy groups through gravitational, dynamic, and magnetized connections. The forces are not literally the same, but the working mode is comparable: field organizations connect, reorganize their binding, exchange energy, and form new stable overall states.

Chemical bond $\leftrightarrow$ scale-relative field coupling.

Within the UQSH framework, chemistry is therefore not reduced, but expanded. It remains the concrete science of atomic and molecular bonds, but is at the same time understood as the expression of a more general field logic: stability arises through suitable coupling, reaction through reorganization, and complexity through the repeated formation of new shared field states.

Chemistry is the flexible coupling of bound field nodes into new stable organizational states.

16.1. Periodic Table and Scale-Relative Organizational Analogy

In established chemistry, the periodic table of the elements describes the ordering of atomic systems according to nuclear charge, electron structure, bonding capacity, and recurring chemical properties. Within the UQSH framework, this order is not replaced. It remains a precise and experimentally confirmed description of atomic matter. The UQSH, however, additionally asks whether the recurrence of stability, coupling, and saturation visible in the periodic table is an expression of a more general working mode of the field.

No literal identification of atoms and galaxies is claimed here. The analogy refers exclusively to functional roles within different scale regimes. What matters is not external sameness, but the question of whether comparable organizational roles recur: a central field node, a binding space, peripheral coupling zones, stable and unstable intermediate states, saturation, neutralization, and reorganization.

In the atomic regime, the nucleus corresponds to the strongly bound center of the field organization. The electron shell describes the region in which coupling, shielding, binding, and chemical reactivity are organized. The position of an element in the periodic table shows which binding roles are stably possible: closed shells, open coupling sites, high reactivity, transitional behavior, or especially stable noble-gas states.

Scale-relatively, a similar logic can also be transferred to larger systems. A galactic nucleus, especially a massive central field node, can assume a nucleus-like function in the superregime. The surrounding disk, the halo, satellite systems, gas structures, and magnetic orders then do not form a literal replacement for electrons, but a peripheral coupling space. In this space it is decided which structures remain bound, which are expelled, which form transitional states, and which forms of reorganization are possible.

The periodic order of atomic elements can therefore be read as an example of a more general UQSH question: Do stable coupling roles recur across scales as soon as a field system forms a center, a binding space, and a limited number of stable external couplings? In this view, the periodic table would not only be a chemical order, but also an indication that field organization favors certain stability patterns.

A scale-relative comparison is:

Atomic nucleus = central binding node [0.3em] Electron shell = peripheral coupling space [0.3em] Valence = open coupling capacity [0.3em] Noble-gas state = closed saturation order [0.3em] Transition elements = flexible reorganization states	(19)
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Transferred to a superregime, this does not yield an identity, but a functional analogy:

Galactic nucleus = central field node [0.3em] Disk and halo = large-scale coupling space [0.3em] Satellites and gas structures = bound external modes [0.3em] Jets and outflows = relief and transition modes [0.3em] Cluster environment = higher-order binding space	(20)
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Chemical and cosmic order can thus be described in a shared language without being identified with one another. In the atomic regime, chemical reactivity corresponds to the capacity to occupy open coupling sites, restructure bindings, or form stable molecules. In the galactic or cosmic regime, comparable roles may occur through accretion, satellite binding, gas inflow, star formation, jet activity, mergers, or cluster coupling. In both cases, the decisive factor is not mass alone, but the way in which a field system carries, limits, shields, or redistributes coupling.

The concept of saturation is especially important here. In the periodic table, certain states appear stable because their outer coupling structure is closed or especially favorably organized. Other states are more reactive because they possess open coupling roles. Scale-relatively, something similar could apply to galaxies: some systems appear compact, quiet, and stabilized; others are in active transitions, showing strong gas dynamics, jets, satellite uptake, star formation, or altered rotation and lensing signatures.

The UQSH prediction is therefore not that a specific element corresponds directly to a specific galaxy class. What would be testable, rather, is whether comparable organizational roles can be found across scales. These include central densification, the number and type of bound external modes, degree of saturation, reactivity, transitional states, neutralization, and coupling capacity. If such roles recur in atomic, chemical, galactic, and cluster-like systems, this would indicate that field reality operates according to similar organizational principles across different scales.

The periodic table therefore remains a chemical order of atomic matter. Within the UQSH framework, however, it is additionally read as a visible example of a deeper field logic: stable reality arises where a center, a binding space, and limited coupling roles form a lasting organization.

The decisive factor is the coupling capacity to the respective regime. Not every existing field configuration must immediately appear as a stable, observable structure. In the subatomic domain, electron-like field configurations could be present without already coupling as electrons of our regime. They would appear as observable states only when their field organization reaches the necessary coupling conditions. Before that, they would not be nothing, but not sufficiently regime-capable.

The same applies scale-relatively to galactic external modes. Dwarf galaxies, gas clouds, dark satellite candidates, or weakly coupled substructures must therefore not simply be equated numerically with the electrons of an atom. For an analogous count, they would have to be distinguished according to their coupling capacity: stably bound external modes, only weakly coupled companion structures, transition modes, unstable reorganization remnants, and field configurations that are not or only barely visible. Only the number of actually coupling-capable roles would be relevant for a scale-relative analogy.

A testable consequence would therefore be that the mere number of satellite structures is not sufficient. What matters would be a coupling-corrected count: Which companions are stably bound, which are only temporarily coupled, which are visible through gas or star formation, which are dynamically present but electromagnetically weak or invisible? Only after such a separation could one investigate whether a galaxy fulfills a specific element-analogous coupling role in the superregime.

Chemical structure and chemical behavior can also be read scale-relatively as indications of recurring field roles. A chemical element is not determined only by its mass or nuclear charge, but by its bonding capacity, reactivity, stability, saturation, and tendency toward transformation. Analogously, a galaxy can also be described not only by mass or size, but by its capacity to bind satellites, absorb gas, form jets, eject matter, activate star formation, or pass into other organizational states.

The comparison with heavy and radioactive elements is especially revealing. A radioactive element such as uranium is not simply a stable endpoint of matter order in the atomic regime, but a heavily loaded, heavy field state that relieves internal tension through decay, radiation, and transformation. Scale-relatively, very massive, overloaded, or highly active galaxies could possess a similar functional role. They would then not simply be larger galaxies, but oversaturated field organizations that relieve their internal tension through jets, outflows, strong radiation, satellite formation, gas redistribution, or activation events.

In this analogy, radioactive decay does not literally correspond to a galactic explosion. What is meant is a shared organizational role: an overly saturated field node cannot hold its tension locally without limit and therefore releases energy, matter, or structuring field motion into its environment. In the atomic regime, this appears as decay and radiation; in the galactic regime, it could appear as jet activity, a quasar phase, a starburst, gas outflow, formation of dwarf galaxies, activation of filaments, or transition into a quieter organizational form.

Conversely, stable chemical elements can serve as an analogy for stable galactic organizational forms. Systems with closed or especially favorable coupling roles behave relatively inertly: they bind their external modes stably, show low reactivity, and change their structure only under stronger external influences. Reactive elements, by contrast, correspond to open field organizations that couple easily, form bindings, or change their outer structure. Scale-relatively, strongly gas-rich, interacting, or satellite-rich galaxies could assume such reactive roles.

Molecule formation also has a possible galactic counterpart. Chemical bonds arise when two or more atomic field organizations align their open coupling roles in such a way that a shared stable structure emerges. On the cosmic scale, galaxy groups, interacting galaxy pairs, cluster cores, or filament nodes could develop similar roles: not every encounter leads to a stable binding, but under suitable velocities, tension states, gas fractions, and coupling conditions, a higher-order organization can arise from several individual systems.

The scale-relative analogy is therefore not:

$$\text{specific element} = \text{specific galaxy}.$$

Rather, it is:

$$\text{chemical behavior} \sim \text{galactic organizational role}.$$

The decisive factors are stability, reactivity, saturation, open coupling sites, transitional states, and forms of relief. A very massive active system may therefore, in the superregime, more closely assume

the role of a heavy, unstable element, while a quiet, closed galaxy corresponds more closely to a saturated or weakly reactive coupling order.

17. Large-Scale Structures and Tension Fronts

If galaxies are field nodes, then their environment is also part of the field configuration. A galaxy does not end sharply at its visible boundary. Its halo, its satellites, its filament connections, and its dynamic excess belong to the overall tension organization. On cosmological scales, the cosmic web can therefore be understood as the visible projection of large-scale field tension.

Filaments, walls, and nodes would then be not only matter distributions, but possible paths, superposition zones, or remnants of propagating tension fronts of higher regimes. From our perspective, such a front would not have to appear as a perfect spherical shell. Because we observe along our light cone and through redshift spaces, a spherical or partly spherical front can appear as an arc, wall, ring, filament chain, or large-scale concentration.

Especially large and partly controversial structures, such as the Hercules–Corona Borealis Great Wall, are interesting in this context as hypothetical test cases. If an extremely large-scale concentration of discharge events or galaxies at similar redshift is present, this could be read within the UQSH framework as the projection of a higher-order activation or tension front. This interpretation remains hypothetical, but it provides a search direction: large-scale structures should be examined not only in terms of matter density, but also in terms of dynamic activation, lensing potential, hot gas, galaxy nodes, and reprocessing indicators.

The nodes of the cosmic web would be especially important here. They could be overlap zones of several fronts. Stronger lensing signals, higher galaxy densities, dynamically active systems, hot gas, and systematic deviations in the relation between baryonic organization and gravitational effect should therefore accumulate there. This connects the large-scale structure of the universe with the same field-tension logic that appears on galactic scales in the dark-matter effect and in q -regimes.

17.1. Imperfect Symmetry as a Prerequisite for Structure

In the idealized limiting case, the spherical tension front has a geometrically clear form. Its surface grows with

$$A(r) = 4\pi r^2.$$

This formula describes the ideal symmetry of a spherical front. In its mathematical ideal form, a sphere is smooth. Physically, however, such a front does not arise in an empty, perfectly homogeneous space, but in an already excited, structured, and coupling-capable field medium.

Even the circle itself shows an important feature in its mathematical description: the ratio of circumference to diameter is determined by π , and π is not a finite, closed number. This does not mean that the mathematical circle is uneven. Rather, it shows that even the simplest ideal round form does not have a fully finite numerical description. Within the UQSH framework, this can be read as an image of the fact that ideal symmetry and real field organization are not identical.

A real tension front passes through a field that already contains motion, gradients, couplings, local tensions, and earlier reorganizations. Its spherical basic form can therefore be preserved without its

surface appearing perfectly homogeneous. Local deviations, phase shifts, density differences, and coupling unevenness are not mere disturbances of a perfect order, but the conditions under which new order can arise at all.

It is precisely the imperfect that creates room for stability formation. If a field front were perfectly smooth, perfectly homogeneous, and without any internal difference, it could hardly form local couplings, resonances, or bound structures. Only small differences in tension, phase, density, and geometry allow certain regions to couple more strongly while others relax or continue to propagate. Structure therefore arises not despite imperfection, but through it.

This also makes clear why large-scale tension fronts in the universe do not have to appear as perfect spherical shells. Their ideal propagation form may be spherical, while their observable realization becomes visible only as arcs, shell remnants, filament edges, rings, or partial fronts. The deviation from the perfect form is not a defect, but an expression of field-mechanical reality: a dynamic medium forms order not through absolute smoothness, but through controlled unevenness.

In this view, logarithmic spiral orders can be understood as relief paths between radial binding and tangential twisting. The golden ratio is not set as a necessary law of every spiral galaxy, but as a possible ideal value of a stable growth relation in which expansion, rotation, and coupling can preserve their form across several scales.

Imperfection is not a disturbance of order, but the condition of its emergence.

18. Planck Limit, Vacuum Energy, and Coupling Filter

Within the UQSH framework, the Planck limit takes on a different meaning. It is not a physical infinity and not necessarily the final structure of the field. It primarily describes the lower resolution and coupling limit of our regime. Below this limit, field motion does not disappear. It continues to reorganize in subregimes that can no longer be directly resolved by our measuring instruments or by our material structure.

This must be distinguished from the upper saturation limit. It appears where local curvature can no longer be further deepened, for example in compact boundary states such as black holes. In short:

**The Planck limit bounds our resolution from below;
field saturation bounds curvature from above.**

This distinction helps with the vacuum-energy problem. In standard physics, an enormous discrepancy arises when zero-point energies are summed up to the Planck scale and compared with the observed cosmological energy density. The difference is often given as about 10^{120} to 10^{122} orders of magnitude. Within the UQSH framework, this number does not have to mean that the calculation captures every real field energy incorrectly. It may also mean that an extremely large amount of field motion lies in subregimes that do not couple directly. The well-known discrepancy between the formally expected vacuum energy and the observed cosmological effect is known as the cosmological constant problem [13, 14].

The decisive distinction is:

$$\rho_{\text{field}}^{\text{total}} \neq \rho_{\text{grav}}^{\text{observed}}.$$

The total field energy includes all nested reorganization and tension components of the Qu-foam. The observed gravitational energy density is only the part that becomes effective and readable as field curvature within our regime. The large discrepancy would then not be proof that the field energy does not exist, but an indication that only an extremely small part of the total organization falls within our gravitational readability.

The vacuum-energy problem thereby becomes a coupling problem:

Not: Why does so much field energy exist?

But: Why does only a small part of it couple into our regime?

One possible structural formula is:

$$\rho_{\text{obs}} = \mathcal{F}_{\text{regime}} \left(S[\rho_{\text{field}}^{\text{total}}] \right),$$

where $\mathcal{F}_{\text{regime}}$ describes the coupling filter of our regime, and S describes saturation. This equation, too, is not meant here as a finished derivation, but as a working form: observed energy density is a filtered and saturated projection of deeper field energy.

18.1. Quantum Fluctuations as Short-Lived Potential Spaces

Within the UQSH framework, quantum fluctuations do not appear as the random emergence of particles from nothing. They arise from the dynamics of the field medium itself. Even below or beyond our direct resonance capacity, field motions, residual tensions, and non-coupling organizational forms remain present.

When such non-coupling field structures encounter one another, overlap, circle around one another, or briefly enter into a shared geometry, a local potential space can arise. For a short time, this potential space has more coherence, more tension, and a higher coupling capacity than its individual components. What was previously not measurable becomes temporarily connectable to our regime in the shared state.

A quantum fluctuation is therefore not an emergence from nothing, but a temporary coupling opening:

$$\begin{aligned} \text{non-coupling field structures} &\rightarrow \text{superposition} \rightarrow \text{short-lived potential space} \\ &\rightarrow \text{measurable fluctuation.} \end{aligned}$$

Scale-relatively, this process can be illustrated through galactic processes. When two galaxies encounter one another, pass through each other, and move apart again, they temporarily generate a larger shared potential space. The individual galaxies remain field nodes of their own, but during the interaction an additional overall organization arises: shared halo tension, shear surfaces, tidal bridges, star-forming regions, shock fronts, and large-scale field distortions. For a hypothetical superregime, this entire process could appear like a short-lived fluctuation: a potential space that appears, couples, and decays again.

The same principle can apply in the subregime. There, field modes that are not directly measurable encounter one another, briefly form a shared order, and then fall apart again. For our regime, this short moment appears as a quantum fluctuation. The fluctuation is then not inexplicable, but the expression of a temporary crossing of the coupling threshold.

The decisive point is that the participating structures do not newly arise from nothing. What is new is only their temporary shared coupling capacity. The fluctuation is the brief becoming-visible of a deeper field motion that was not readable individually, but that together forms a measurable potential space.

Quantum fluctuations are short-lived coupling windows of deeper field organization.

19. Electromagnetism as Twisting Order of the Field

Within the UQSH framework, electromagnetism is understood as the directed twisting order of the field medium. The starting point is the assumption that the field, in its undisturbed state, has no preferred direction. Free field motion can propagate spherically; bound field motion can stabilize as matter, mass, and gravitation. Direction arises only where field motion is locally superposed, bound, twisted, polarized, or coherently aligned.

Within the UQSH framework, gravitation is not reduced to bound matter or mass. It arises wherever field tension forms an effective curvature within a regime. Planets, stars, galaxies, halos, filaments, tension fronts, and saturated nodes are therefore different carriers of gravitational field response insofar as their curvature couples into the respective observational regime.

In this view, magnetism does not describe a second fundamental substance alongside gravitation. It arises when a field curvature is not only effective, but coherently twisted, polarized, and directed. While the gravitational response describes the regime-compatible curvature effect of a field state, magnetism appears as the short-range, axially ordered near projection of this curvature.

Gravitation = regime-effective field curvature,

Magnetism = coherently twisted near-order of this field curvature.

This interpretation explains why magnetic action can be strong, local, and direction-dependent. In a magnet, it is not the entire mass that couples uniformly outward, but a part of the internal field organization that emerges as a coherent twisting order. This order can be read as a directed near curvature of the field. Its strength depends on how many internal field modes are aligned, how stable this alignment remains, and how well it couples into the surrounding field medium.

A magnet is therefore not a perfectly aligned body. It contains many subregime twisting modes in different directions. Macroscopic magnetization arises when one particular twisting direction statistically and coherently predominates. Other twisting components remain present. They can generate internal tension fields, residual couplings, hysteresis, heat losses, or electric voltage differences.

macroscopic magnetization = dominant coherent twisting direction,

residual tensions = not fully aligned twisting components.

This makes it understandable why different materials respond differently to magnetism. In some substances, the internal field twistings largely cancel one another. They have only a weak or short-lived magnetic response. In other substances, internal twisting modes can be aligned, but fall back again once the external field is removed. In ferromagnetic and permanent-magnetic materials, by contrast, many of these modes can align collectively and remain stable. The magnetic strength of a material therefore describes its capacity to project subregime field twisting coherently into our regime.

The behavior of a divided magnet also follows from this. When a magnet is broken, north and south poles are not separated from one another as distinct substances. What is divided is an internal twisting order. Each piece forms a complete polarization axis again from its own field organization. This is why each fragment again has a north and a south pole. Polarity belongs not only to the ends, but to the continuous internal alignment of the field node.

In this view, electricity appears as the open tension form of the same twisting logic. While magnetism describes a bound or closed twisting order, electrical action denotes an open coupling asymmetry or tension difference. When the magnetic order is shifted, changed, or relieved, electric tensions can arise from the twisting components that are not fully balanced. In this sense, electricity is not separate from magnetism, but the moving or released tension side of the same field twisting.

magnetic order = stored directed field twisting,

Electricity = open or moving tension difference of this twisting.

Electromagnetism therefore denotes the dynamic interplay between bound twisting and open tension. A change in the magnetic twisting order can produce electrical field work; an electrical tension motion can in turn build magnetic twisting. What appears in the standard picture as the coupling of electric and magnetic fields is read, within the UQSH framework, as reciprocal conversion between open coupling tension and bound twisting order.

This view can be extended scale-relatively. Planets, stars, magnetars, accretion disks, galaxies, and filaments are not only accumulations of mass, but rotating, layered, or flowing field nodes. Where such systems possess enough internal motion, plasma, charge separation, shear, or coherent twisting, they can form large-scale magnetic orders. Magnetism then acts not only as a small material property, but as an axial, channeling, and stabilizing function in the larger field.

Earth, for example, possesses not only an overall gravitational tension, but also a directed planetary field order. A compass does not align abstractly with the North Pole; rather, it couples its own internal twisting axis to the larger magnetic axial order of Earth. In the same way, solar magnetic fields can order plasmas, solar wind, flares, and coronal structures. Magnetars show an extreme form of saturated magnetic twisting. In spiral galaxies, magnetic field orders can help shape gas flows, star formation, plasma streams, and spiral-arm stability.

This gives two complementary readings of the same field basis. Gravitation describes the regime-effective curvature tension of a field state. Magnetism describes the part of this field order that is coherently twisted, axially polarized, and thereby locally directed in its effect. The general saturation limits of gravitation were treated above; what matters here is that magnetism does not deepen this field

effectiveness, but directs it.

This produces poles, channels, relief directions, field lines, plasma streams, and preferred coupling axes. Gravitation gives the field state its curving effectiveness; magnetism makes part of this effectiveness locally directed and coupling-selective.

field curvature $\rightarrow$ regime-effective gravitation,

coherently twisted field curvature $\rightarrow$ magnetic near-order.

This interpretation also has a speculative technological consequence. If magnetism is a directed twisting form of field tension, then partial decoupling from the field gravitation of a larger body would not be expected through switching off gravitation. Rather, it would have to arise through a controlled reorganization of one's own coupling twist. A body would have to alter its local field attachment in such a way that it is not merely passively located in the larger gravitational field, but redirects its embedding in a directed way.

An ordinary body has passive field binding. A magnet has directed field binding. An electromagnet has controllable directed field binding. A more advanced field technology would have to extend this control to the entire body: not merely generating a magnetic field, but a coherent, dynamically controlled twisting geometry of the whole field node.

passive mass $\rightarrow$ bound field tension,

Magnet $\rightarrow$ directed field tension,

field coupling system $\rightarrow$ controlled twisting geometry.

In this context, reports of rotating or disk-shaped objects should be understood only as speculative illustration, not as evidence. If a technical system could modulate its coupling to the surrounding field medium, the whole body would not necessarily have to rotate mechanically. An internal rotation would also be possible: a circulating ring, a plasma layer, a field rotor, a superconducting current, or a phase-stable coupling zone. The decisive factor would not be the visible rotation of a shell, but the rotation of coherent field tension inside it.

Such a circulating twisting order could generate a directed counter-order or transverse order relative to normal gravitational embedding. The object would then not simply accelerate against its inertia, but change its coupling to the larger field space. In this view, motion would not only be force against mass, but a change in field attachment.

This outlook remains highly speculative. It does, however, show what role magnetism could play in the UQSH in principle: it is not merely a side effect of electric charges, but the first observable indication that matter can carry its field tension not only passively, but organize it directionally. The next stage would not be stronger magnetization alone, but the controlled, dynamic, and scale-compatible steering of this twisting order.

Every regime-effective field curvature forms gravitation;
coherent twisting forms its magnetic near-order.

Field decoupling would not be a switch-off of gravitation,
but a controlled redirection of the coupling twist.

20. The Four Interactions within the UQSH Framework

In this view, the four known interactions do not appear as four ontologically separate forces. They are four coupling modes of the same field medium. The difference lies in the degree of organization, in range, in saturation, and in the freedom of reorganization.

20.1. *Gravitation*

Gravitation is the universal field response to organized tension curvature. Every stable or metastable field organization changes the tension state of the medium. For bound observers, this change appears as gravitation. Gravitation is not the maximally locally bound saturation mode, but the large-scale continuation of that field curvature which is not fully locked locally. This is why it is locally weak, but universal and not shieldable.

20.2. *Electromagnetism*

Electromagnetism was previously described as the interplay between open coupling tension and bound twisting order. In the systematics of the four interactions, it forms the directed, regime-internal coupling layer between free radiation, bound matter, charge asymmetry, magnetism, and chemical organization.

20.3. *Strong Interaction*

The strong interaction is the local saturation mode of maximally densified field nodes. In a nuclear regime, the field organization is so densely nested that deviations cannot freely run outward. They are immediately returned to the existing structure and bound into it. Its strength arises from this local locking. Its short range is the reverse side of the same property: because the tension remains almost completely locally bound, it can hardly form a long-range signature outward.

Scale-relatively, this can be compared with the difference between an atomic nucleus and a galactic center. In the atomic nucleus, the binding is so dense that protons and neutrons do not appear as freely separating field nodes, but remain locked within a shared nuclear order. On the galactic level, the central field node, such as a dense core region or an Anun-near state, assumes a comparable organizational role. There, too, field motion is strongly bound locally, stabilizes the surrounding structure, and allows only limited relief outward. Jets, accretion zones, or boundary-field reliefs would then not be a direct counterpart of the strong interaction, but macroscopic escape forms that arise when local locking reaches its saturation limit.

20.4. *Weak Interaction*

The weak interaction is not a permanently binding mode, but a selective reorganization channel of boundary-stable field patterns. It emerges where an existing configuration can no longer stably carry its old order and passes into another allowed form. Decay and transformation therefore appear not as the destruction of particles, but as transitions between field patterns.

Strong interaction	=	locally saturated binding of dense field nodes
Electromagnetism	=	oriented modulation coupling of bound nodes
Weak interaction	=	selective reorganization of boundary-stable patterns
Gravitation	=	large-scale field response to tension curvature

The fine-structure constant α is not yet treated formally here. It remains a separate next step, because it presumably expresses the dimensionless relation between electromagnetic modulation coupling, bound orientation asymmetry, and the maximum reorganization rate.

Scale-relatively, this can be compared with transition states between stable orders. In the atomic nucleus, the weak interaction appears where a neutron-like or proton-like state can no longer stably maintain its previous coupling role and passes into another allowed role. On the galactic level, analogous processes would not be direct particle decays, but slow reorganization processes of boundary-stable systems: an unstable satellite node can be incorporated into a larger galaxy, a gas or star-forming region can tip into a new stable structure, or a small field node can lose its former order through tidal forces, jets, shock fronts, or relief zones and pass into another coupling form.

The analogy therefore does not lie in a literal identification of beta decay and galactic development, but in the shared functional role: a boundary-stable pattern changes its state because the old order is no longer sustainable. In the microphysical regime, the weak interaction is the precise, quantized channel of such transformation; on larger scales, the same logic appears as a slow, structure-dependent reorganization of field nodes.

21. **Recurring Field Working Mode and the Possibility of Lasting Life Formation**

A central motif of the UQSH is the assumption that the field medium works according to the same basic principles across all dimensional scales. Radiation, matter, gravitation, dedensification, densification, saturation, relaxation, and reorganization are not separate processes, but different expressions of the same field working mode. The field moves, tensions, binds, relieves itself, and continuously reorganizes its states.

This leads to a different picture of cosmic development. Life is then not the result of a one-time, unrepeatable accident in an otherwise passive universe, but a possible consequence of recurring organizational chains. Because neither complete saturation nor complete relaxation is ever actually reached, the field remains in residual dynamics, tension, relief, and reorganization. As a result, new star-forming regions, new forms of matter, new chemical environments, and new coupling windows arise again and again.

The prerequisite for this is not that the same conditions occur identically everywhere. What matters is that the field repeats the same working logic: free field motion becomes bound, bound organization produces gravitation, gravitation collects matter, matter forms stars, stars produce heavier elements, elements enable chemistry, chemistry produces complex coupling networks, and coupling networks can, under suitable conditions, develop biological organization.

field motion → matter formation → stars → elements
→ chemistry → complex organization → life.

In this view, life is not foreign to the universe. It is an especially complex, locally stabilized, and self-reorganizing field organization. Biological systems do not exist outside field mechanics, but are late coupling forms of the same dynamics that also carries radiation, matter, gravitation, and chemical bonding. Life is therefore not separate from physics, but a higher form of readability of organized field motion.

The boundary regions are especially important here. In filaments, halos, accretion zones, star-forming regions, planetary transition zones, and Anun relief regions, densification and relaxation meet. Such zones are not only places of high dynamics, but also places of selective stabilization. There, field motion can pass into new bound forms, matter and radiation can be modulated again, and the conditions for chemical and biological complexity can arise.

This also changes the role of destruction. Supernovae, collisions, jets, radiation bursts, accretion, and galactic reorganization are not only destructive events. They distribute field motion, heavy elements, dust, plasma, and new coupling conditions. What appears locally as an end can elsewhere become the prerequisite for new order. Relief and decay are therefore not merely opponents of life, but part of the cosmic redistribution through which later habitats become possible.

decay → redistribution → new coupling conditions → new organization.

From this view follows a natural prediction: if water can form and remain stable wherever hydrogen, oxygen, suitable temperatures, pressure conditions, and chemical coupling spaces come together, then life-capable environments are not in principle limited to a single place. Life is not therefore expected automatically everywhere, but its emergence is understood as a repeatable field process: where water remains stable permanently or cyclically, where chemical gradients exist, where energy flow is present, and where there is enough time for self-organization, life-like or recognizable forms of life can arise.

Such life forms develop differently depending on their environment. On water-rich planets, in subsurface oceans, in icy worlds, in hydrothermal systems, in dense atmospheres, or at interfaces between rock, water, ice, plasma, and radiation, different coupling conditions can prevail. Life is therefore not a fixed form, but purpose-bound self-organization within a local coupling window: its form follows the environment, because every living being derives its stability from the field, energy, water, temperature, and matter conditions present there.

water + energy flow + chemical coupling + temporal stability → life-capable self-organization.

The UQSH thereby broadens the question of life in the universe: the decisive question is not only

whether individual planets have favorable conditions, but whether the field medium repeatedly generates coupling windows in which order, energy flow, water chemistry, and stability work together.

Where water, energy flow, and stable coupling windows exist,
the field can repeatedly produce life-capable organization.

22. Formalization Program

The present work remains largely conceptual in its ontological interpretation. It does not, however, stand outside established mathematical formalizations. Many central relations of physics, such as $E = mc^2$, the Newtonian approximation of gravitation, the relation between frequency and energy, or the invariance of the speed of light, are not rejected within the UQSH framework. Instead, they are read as already measured limiting and coupling forms of a deeper field order.

The formalization program therefore does not consist in reinventing all known equations, but in working out their shared field-mechanical interpretation. Established formulas can be understood as limiting cases, regime projections, or effective descriptions of Q , Φ , and $B(\Phi)$. For a further mathematical development, several basic quantities must be clearly distinguished:

Q Qu-ground; the deepest ontological foundation of the UQSH.

Φ Physically readable field projection of Q .

$B(\Phi)$ Motion of this field projection.

$K^{(n)}(x, t)$ Field curvature or tension curvature of a regime n .

$M^{(n)}(x, t)$ Modulation structure within a carrier front.

$S_n(K)$ Saturation function of the regime.

$\mathcal{F}_n$ Coupling filter between field organization and observational regime.

R_n Reorganization rate, whose limiting value appears in our regime as c .

In the case of Φ , the term *projection* does not mean virtuality or mere appearance. It means the real field form in which Q becomes physically accessible for a regime.

The fundamental UQSH order is therefore formulated as an ontological and dynamic layering:

$$\boxed{Q \rightarrow \Phi \rightarrow B(\Phi)} \quad (21)$$

Here, Q denotes the Qu-ground, Φ its physically readable field projection, and $B(\Phi)$ the motion of this field projection. From this motion, the first physical readings arise:

$$\boxed{B(\Phi) \Rightarrow L, , E, , \mathfrak{G}} \quad (22)$$

Here, L denotes the freely propagating form of motion of the field projection, E its motion content or its timing and frequency order, and $\mathfrak{G}$ its regime-effective curvature and tension aspect. Light and

gravitation are therefore not equated. They appear as different readings of the same underlying field motion.

A simple structural form for observable modulations would be:

$$\text{observed modulation} = \mathcal{F}_n \left[M^{(n)} \left(K^{(n)}, K^{(n-1)}, K^{(n+1)} \right) \right]. \quad (23)$$

For redshift, a dedensification function would have to be formulated:

$$\nu_{\text{obs}} = \nu_{\text{em}}, E_{\text{path}}^{-1}, \quad (24)$$

where E_{path} describes the cumulative field-mechanical dedensification along the propagation path.

For gravitation, it would accordingly have to be tested whether the observed excess can be written as a saturated projection of bound and free field tension:

$$g_{\text{obs}} = g_{\text{bar}} + \mathcal{F} * \text{gal} [S_n (K * \text{bind} + K_{\text{free}})] . \quad (25)$$

Here, K_{bind} describes the tension curvature of bound field organization, such as baryonic matter, while K_{free} describes the tension curvature of freely propagating tension fronts. Free light-tension fronts therefore do not stand outside field reality. They are free forms of motion of Φ and carry timing, tension, and curvature relation. Their gravitational effectiveness, however, is not identical with classical gravitation, but the regime-dependently readable curvature aspect of the same field motion.

The classical Newtonian limiting case is preserved when the filtered field tension in the weak, quasi-static regime is read as effective energy or mass:

$$g(r) = G_N \frac{E_{\text{eff}}}{c^2 r^2}. \quad (26)$$

Here, G_N is not gravitation itself, but the empirical coupling scale with which our regime measures field tension as acceleration.

These equations are not meant as a finished theory. They show which building blocks are required: Qu-ground, field projection, field motion, saturation, coupling, modulation, dedensification, free and bound field tension, and regime projection.

A concrete application of this program is the energetic regime width between Anektron and Anun.

22.1. Preliminary Derivation of the Regime Width between Anektron and Anun

The terms *Anektron* and *Anun* allow a first formalizable description of the energetic span of a scale-relative regime. A regime (R_n) is not defined by an arbitrary size scale, but by two saturation limits: a minimal saturation form from which an independent coupling unit becomes measurable, and a maximal saturation form beyond which local deepening can no longer be continued.

As defined above, the *Anektron* denotes the lower saturation limit of a regime, while the *Anun* describes the upper saturation limit. For the formalization, the decisive point is that both boundary forms can be treated as characteristic energies of the same regime.

For a regime R_n , we therefore define two characteristic energies:

$$E_A^{(n)}$$

as the energy of the minimally saturated Anektron form and

$$E_U^{(n)}$$

as the energy of the maximally saturated Anun form.

The energetic span of a regime is then

$$\Lambda_n = \frac{E_U^{(n)}}{E_A^{(n)}}.$$

This quantity Λ_n does not simply describe a mass ratio, but the linear saturation span of a regime: the distance between the first measurable saturated coupling and maximum local field saturation.

For our observational regime, the electron rest energy can be used as a first calibration point for the lower coupling boundary:

$$E_A^{(n)} \approx E_e = m_e c^2.$$

This does not mean that electron and Anektron are fully equated, but that the electron energy serves as an observable approximation of the minimal coupling form.

As the upper calibration, the minimal black-hole boundary can be used. In this view, a black hole is the strongest candidate for an Anun of our regime:

$$E_U^{(n)} \approx M_{\text{BH},\text{min}} c^2.$$

It follows for the regime width that

$$\Lambda_n = \frac{M_{\text{BH},\text{min}} c^2}{m_e c^2} = \frac{M_{\text{BH},\text{min}}}{m_e}.$$

The speed of light cancels out. In this first approach, the regime width is therefore a pure mass or energy ratio between lower minimal coupling and upper maximum saturation.

If one takes the minimal black-hole boundary to be approximately a few solar masses,

$$M_{\text{BH},\text{min}} \sim 2 - 5 M_\odot,$$

then Λ_n lies, as an order of magnitude, at about 10^{61} . For the commonly used calibration value

$$M_{\text{BH},\text{min}} \approx 3 M_\odot$$

one obtains

$$\Lambda_n \approx \frac{3M_\odot}{m_e} \approx 6.5 \times 10^{60}.$$

The quadratic span is especially interesting:

$$\Lambda_n^2 \sim 10^{122}.$$

This order of magnitude lies in the range of the known discrepancy between the formally expected vacuum energy and the observed gravitational effect, and motivates the following working hypothesis:

$$\frac{\rho_{\text{vacuum}}^{\text{formal}}}{\rho_{\text{grav}}^{\text{observed}}} \sim \Lambda_n^2.$$

Here, $\rho_{\text{vacuum}}^{\text{formal}}$ denotes the vacuum energy density formally expected from field fluctuations, and $\rho_{\text{grav}}^{\text{observed}}$ denotes the actually observable gravitational effect. The quadratic dependence is not claimed here as a final derivation, but as an Ansatz: if coupling between regimes acts not linearly, but projectively, surface-like, or through two-sided filtering, the observable discrepancy may appear as a quadratic trace of the regime width.

This gives rise to a UQSH interpretation of the vacuum-energy discrepancy: not all formal field energy has to become gravitationally effective in our regime. Only a filtered part of the nested field organization couples as observable gravitation. The known discrepancy would then not be an expression of too much energy, but of too little direct coupling between total field motion and our observational regime.

The regime relation can therefore be summarized as

$$\rho_{\text{field}}^{\text{total}} \gg \rho_{\text{vacuum}}^{\text{formal}} \gg \rho_{\text{grav}}^{\text{observed}}.$$

The total field motion may be substantially larger than the vacuum energy formally calculated up to the Planck limit. This, in turn, may be substantially larger than the gravitational projection actually observed. The reason lies in saturation, coupling filtering, and regime limitation.

The regime transition itself can be written as a recursive relation:

$$E_A^{(n+1)} \approx E_U^{(n)}.$$

This means: the Anun of one regime can assume the function of an Anektron in the next higher regime. The end of maximum saturation in one regime thereby becomes the beginning of minimal coupling capacity in the next regime.

This gives the scale-relative chain:

$$E_A^{(n)} \rightarrow E_U^{(n)} = E_A^{(n+1)} \rightarrow E_U^{(n+1)}.$$

In words: a regime begins with minimal saturation, unfolds its matter and structure forms within these

limits, reaches maximum saturation in its Anun, and can precisely thereby be translated into the next higher regime.

The relation developed here is not a completed mathematical derivation of the electron, the black hole, or the cosmological constant. It does, however, represent a grounded preliminary derivation. It connects three quantities that have so far been treated separately within a shared saturation framework:

Electron/Anelectron, Black hole/Anun, vacuum-energy discrepancy.

The central UQSH Ansatz is therefore:

$$\Lambda_n = \frac{E_{\text{Anun}}^{(n)}}{E_{\text{Anelectron}}^{(n)}} \sim 10^{61}$$

and

$$\Lambda_n^2 \sim 10^{122}.$$

The first quantity describes the linear energetic width of our regime. The second quantity could represent the quadratic coupling or projection trace of this regime width in the observed vacuum-energy discrepancy.

23. Soft Anun–Anelectron Transition Band and Temporal Resolution

Within the UQSH framework, the minimal black-hole mass does not automatically mark a fully developed Anelectron form of the next higher regime. It first denotes the lower threshold from which a field state of our regime reaches the Anun function. From this point on, the local curvature depth is saturated. Further field motion does not deepen the state without limit, but increases range, boundary structure, coupling surface, and inertia.

This does not produce a hard point-like regime transition, but a soft transition band. A minimal Anun may initially be only weakly, fleetingly, or not yet coherently readable for the superregime. With increasing mass and range, however, the saturated field form becomes increasingly coarser, more inert, and more stable. It can thereby gradually enter the observational and coupling spectrum of the next higher regime.

This readability depends not only on the mass or energy of the Anun, but also on the temporal resolution of the observing regime. A process that unfolds over long periods within our regime may appear to a superregime as a short or soft transition. What appears to us as the slow growth of a black hole through accretion, merger, and field saturation could be read in the superregime as the emergence of a new stable coupling unit.

The regime bridge is therefore temporally relative. The lower threshold of the transition is given by minimal Anun formation,

$$E_{\text{U,min}}^{(n)} \sim M_{\text{BH,min}} c^2.$$

Coherent readability in the next higher regime, however, is reached only in an extended transition

range:

$$E_A^{(n+1)} \in \left[E_{U,\min}^{(n)}, E_{U,\text{coh}}^{(n)} \right],$$

where $E_{U,\text{coh}}^{(n)}$ denotes the Anun energy from which the saturated field form becomes stably readable for the superregime as a coherent Anektron function.

This transition can occur very rapidly dynamically if the minimal Anun quickly binds further field motion. In our regime, this may appear as strong accretion growth, merger, jet activity, or increasing boundary-field relief. In the superregime, the same process would not be visible as the sum of individual fast processes, but as the increasing coherence of a new coupling unit.

Temporal resolution is therefore decisive. A regime does not read every subordinate dynamic individually, but only the development that becomes coherent within its own time measure. An Anun of our regime can thus initially lie below superregime readability, but then pass through rapid growth into a stable Anektron function of the superregime.

In short: the threshold of regime transition is set by minimal saturation, but the observability of the new Anektron arises softly, time-dependently, and coherence-dependently.

Saturation sets the threshold; temporal coherence makes the transition observable.

An Anun does not fully become the Anektron of the next higher regime at the same instant. It grows into its observability.

24. Astrophysical Mass Gap as a Saturation Zone

In the astrophysical context, the so-called mass gap, that is, an observed mass gap between compact stellar remnants and black holes, can be understood within the UQSH framework as a transition region between the maximum carrying capacity of matter organization and the minimally stable formation of an Anun. A compact star can carry its field organization only up to a certain limit. If this limit is exceeded, the existing matter structure is no longer sufficient to hold the local field tension stably. The field has to reorganize. However, this does not necessarily produce a fully coherent Anun form immediately. An unstable saturation zone may therefore lie between the two states.

Formally, this region can be written as

$$M_{\max,\text{matter}}^{(n)} < M < M_{\text{Anun},\min}^{(n)}.$$

Here, $M_{\max,\text{matter}}^{(n)}$ denotes the maximum mass that can still be carried as compact matter organization in the regime R_n , while $M_{\text{Anun},\min}^{(n)}$ denotes the minimum mass from which a lasting Anun function becomes possible. In this view, the mass gap would not be an empty region in an absolute sense, but an unstable reorganization region: too strongly densified for ordinary matter stability, but not yet fully coherent as an upper saturation form.

**The astrophysical mass gap is the unstable zone
between matter carrying capacity and Anun coupling.**

With increasing mass, range, and boundary structure, such a state can enter the Anun transition band. The minimal black-hole mass then marks not the end of all dynamics, but the first clear threshold from which maximum saturation can appear as the upper regime form.

24.1. Yang–Mills Mass Gap in the Light of the UQSH

The Yang–Mills mass gap problem is not solved mathematically here, but only interpreted from the perspective of the UQSH. In its usual formulation, the question is that a quantized non-abelian gauge theory has a positive gap between the vacuum and the first physical massive excitation state, despite having massless fundamental fields. Schematically, this can be written as an energy gap:

$$\Delta E_{\text{YM}} = E_1 - E_0 > 0.$$

Within the UQSH framework, this gap can be read as the expression of a minimal saturation threshold. Free field motion or massless fundamental fluctuation does not automatically become observable as an independent inert state. Only when the field organization crosses a minimal binding and saturation threshold does a stable, mass-bearing, coupling-capable mode arise.

The simplest UQSH reading first formulates this threshold through the saturation condition:

$$E_{\text{A}}^{(n)} = \min \left\{ E \mid S_n(K) \geq S_{\text{min}}^{(n)} \right\}.$$

The coherence condition added later further refines this threshold. Here, $E_{\text{A}}^{(n)}$ denotes the minimal Anelectron energy of a regime, $S_n(K)$ the saturation function of the field curvature or field organization K , and $S_{\text{min}}^{(n)}$ the minimal saturation threshold from which an independent inert coupling mode can arise. In this reading, the Yang–Mills mass gap appears not as an additional property beside field motion, but as the formal trace of the threshold at which free field organization first passes into measurable inertia.

In the UQSH, the Yang–Mills mass gap corresponds to the minimal saturation threshold between free field motion and an inert coupling form.

This interpretation does not replace a mathematical Yang–Mills derivation. It merely places the problem into the language of the UQSH: mass appears there not as an isolated substance, but as the result of minimal saturation, binding, and regime-capable stabilization.

24.2. Coherence, Hysteresis, and Observational Filters

High energy or field tension alone is not sufficient for the formation of stable field forms. The field organization must be sufficiently coherent in space, time, and geometry in order to couple as an independent mode. At the same time, saturation has a hysteresis: once a saturated state has been reached, it can persist even if the original excitation partly decreases. Finally, each observational regime reads such states only through its own filter of energy, time, and structural resolution. These three conditions are formally added in what follows.

24.3. Formal Addition: Coherence, Hysteresis, and Observational Filters

High energy or field tension alone is not sufficient for the coupling capacity of a field state. A state must also be sufficiently coherent in order to appear as an independent field form within a regime. We therefore introduce a coherence function

$$C_n(x, t) \in [0, 1]$$

for a regime R_n . It describes how strongly a field state is ordered spatially, temporally, and geometrically. Formally, it can be written as the product of three contributions:

$$C_n = C_{\text{space}}^{(n)} C_{\text{time}}^{(n)} C_{\text{phase}}^{(n)}.$$

Here, $C_{\text{space}}^{(n)}$ describes the spatial stability of the field form, $C_{\text{time}}^{(n)}$ its duration relative to the time resolution of the regime, and $C_{\text{phase}}^{(n)}$ its internal phase or geometry order.

A field state becomes readable as its own coupling form only when both its saturation and its coherence exceed a threshold:

$$S_n(K) \geq S_{\min}^{(n)} \quad \text{and} \quad C_n \geq C_{\text{crit}}^{(n)}.$$

This gives, for minimal Anektron formation:

$$E_A^{(n)} = \min \left\{ E \mid S_n(K) \geq S_{\min}^{(n)} \wedge C_n \geq C_{\text{crit}}^{(n)} \right\}.$$

For upper Anun formation, correspondingly:

$$E_U^{(n)} = \min \left\{ E \mid S_n(K) \geq S_{\max}^{(n)} \wedge C_n \geq C_U^{(n)} \right\}.$$

Here, $S_n(K)$ denotes the saturation function of the field curvature or field organization K , while $C_U^{(n)}$ describes the coherence required for a maximally saturated state not merely to arise transiently, but to be carried as a stable Anun form.

Saturation also has hysteresis. Entering a saturated state and leaving it do not necessarily occur at the same threshold. We therefore distinguish an activation threshold $S_{\text{on}}^{(n)}$ and a maintenance threshold $S_{\text{off}}^{(n)}$:

$$S_{\text{on}}^{(n)} > S_{\text{off}}^{(n)}.$$

A state can thus remain stable even if the original saturation excitation has already decreased somewhat:

$$S_n(K) \geq S_{\text{off}}^{(n)} \quad \Rightarrow \quad \text{saturated form remains preserved.}$$

This hysteresis explains the longevity of stable field forms. Once formed, inertia does not dissolve immediately as soon as the external excitation decreases. The field organization has a memory of its saturation.

For observation by a regime R_n , an additional coupling filter can be introduced:

$$\mathcal{F}_n = \mathcal{F}_E^{(n)} \mathcal{F}_t^{(n)} \mathcal{F}_L^{(n)}.$$

Here, $\mathcal{F}_E^{(n)}$ describes the energy resolution, $\mathcal{F}_t^{(n)}$ the time resolution, and $\mathcal{F}_L^{(n)}$ the structural or length resolution of the observational regime.

The observable field effect of a state then does not arise from the total field energy, but from the filtered, coherent, and saturation-weighted projection:

$$W_{\text{obs}}^{(n)} = \mathcal{F}_n [C_n S_n(K) E_{\text{field}}] .$$

A field state appears as its own object when

$$W_{\text{obs}}^{(n)} \geq W_{\text{crit}}^{(n)} .$$

This condition connects energy, coherence, saturation, and observability. It shows why not every fluctuation forms matter, why not every large mass immediately possesses Anun function, and why not every existing field organization is measured as an object in its own right.

In summary:

Coupling = filtered effect of energy, coherence, and saturation.

Or formally:

$$W_{\text{obs}}^{(n)} = \mathcal{F}_n [C_n S_n(K) E_{\text{field}}] \geq W_{\text{crit}}^{(n)} .$$

In simple terms, this means:

A regime does not see the whole field energy. It sees only the part that is sufficiently saturated, sufficiently coherent, and readable through the observational filter of that regime.

25. Predictions and Testable Consequences

From the UQSH perspective, several testable expectations follow. They should not be understood as already proven statements, but as a working program: if the UQSH describes physical field reality correctly, then recurring couplings between light, spectrum, gravitation, matter organization, redshift, and large-scale field structure should appear in existing and future data.

Galactic Dynamics and the Dark-Matter Effect

- The dark-matter effect should be more closely connected to the spatial organization of baryonic structures than to an independent halo component that can be distributed arbitrarily.
- Galaxies with similar total mass but different internal organization should show different excess profiles if their field nodes are activated to different degrees.
- The dynamic factor D_{DM} should not grow arbitrarily with radius, but should point in large systems to a field-mechanical saturation or instability limit.
- In massive spiral galaxies, more diffuse regimes should dominate because the field tension is distributed over larger scales and smoothed more strongly.

- The Radial Acceleration Relation should appear not only as an empirical correlation, but should contain residual structures that point to different field-organization regimes.
- Core-cusp deviations should be systematically connected to saturation, gas dynamics, star formation, radiation pressure, and the reorganization history of the respective galaxy.
- Galactic rotation should be explainable not only from present mass, but from the history of field tension, activation events, and relaxation states.

Lensing, Clusters, and Large-Scale Field Tension

- Gravitational lenses should not always follow the baryonic mass exactly, but the most stable and coherent field-tension organization.
- In colliding clusters, lensing centers should preferably lie where coherent galactic field nodes remain preserved, while hot gas loses part of its coherent field effect.
- Systems such as the Bullet Cluster should not be read as special cases, but as especially visible examples of a separation between baryonic turbulence and coherent field tension.
- The strength of the lensing signal should be connected to the organizational history of a cluster, not only to the currently visible matter distribution.
- Clusters with many stable substructures should show stronger collective field tension than similarly massive but more strongly relaxed or more turbulent systems.
- Large-scale structures should be examined not only in terms of matter density, but together with lensing, hot gas, activation events, filament connections, and dynamic excess.
- In filaments, a weak but large-scale coherent field-tension effect should be detectable that does not follow entirely from visible mass alone.
- Lensing maps and galactic dynamics should be combinable more effectively through shared field-tension parameters than through purely local mass densities.

Light, Spectrum, and Electromagnetic Modulation

- Galactic spectral templates should retain distinguishable organizational classes even after strong smoothing, if the coarse field-node signature is real.
- Spectral lines should be readable not only as properties of individual atoms or molecules, but also as traces of the modulation history of the carrying tension front.
- Combinations of electromagnetic modulation and gravitational field response should provide more information than either level separately.
- Strongly radiating systems should, over time, produce a measurable field-mechanical pre-structuring of their environment, especially in active galactic nuclei, quasars, star-forming regions, and jets.

- Different types of radiation should not appear as fundamentally different kinds of motion, but as different degrees of timing, tension, and modulation of the same free field motion.
- High-frequency radiation should correlate more strongly with local reorganization processes, matter transformation, and field activation than low-frequency radiation that is already more strongly dedensified.
- Absorption and re-emission should be describable not only as the exchange of discrete photons, but as the local reorganization of bound field nodes.
- The wave–particle dualism should continue to be experimentally readable as the difference between extended front propagation and local coupling to detectors.

Redshift, Voids, and the Dark-Energy Effect

- Cosmological redshift should contain at least partly path-dependent components if propagating tension fronts dedensify field-mechanically during their propagation.
- Lines of sight through large voids should show systematically different redshift and smoothing effects than lines of sight through filament-rich or cluster-rich regions.
- Path-dependent redshift effects should be partly averaged over very large distances and thereby become harder to detect. They would be more clearly testable for well-determined, comparable lines of sight that, in close relation to one another, pass once through voids and once through filament-rich, more strongly tensioned, or cluster-adjacent field regions.
- Voids should not appear as the main cause of the dark-energy effect, but as regions in which dedensification proceeds especially undisturbed and therefore especially clearly.
- The dark-energy effect should be more strongly connected to large-scale dedensification of propagating field motion than to an independent substance or constant external force.
- The measured redshift of very distant objects should contain, in addition to distance, indications of the field regimes through which the radiation has passed.
- Particularly old or very distant radiation should show increasing smoothing and a decrease in internal organizational density, without this necessarily having to be interpreted as pure expansion of space.
- Within the UQSH framework, the visible boundary of the universe should remain testable as a coupling and range boundary of our observational regime, not necessarily as an absolute boundary of physical reality.
- Path-dependent redshift and smoothing effects should be especially testable in the nearer cosmic environment where the distances of the objects under consideration are not derived exclusively from redshift. Lines of sight through voids and lines of sight through more strongly tensioned field regions, such as filaments or cluster-adjacent regions, could then show different effective radiation temperatures or degrees of spectral smoothing. Within the UQSH framework, voids would be expected to allow a freer and less disturbed dedensification of the tension front. Radiation fronts along such paths could therefore appear slightly less cooled, or relatively

higher in temperature, than comparable lines of sight through more strongly tensioned, more coupling-rich field configurations.

CMB and Large-Scale Background Structure

- CMB anisotropies should partly reflect path-dependent and large-scale field-tension states, especially in connection with voids, filaments, nodes, and cluster axes.
- The CMB should be read not only as a relic state, but also as a long-range smoothed coupling band of propagating field motion.
- CMB cold and warm regions could correlate with large-scale relaxation, dedensification, or tension regions.
- CMB anomalies, such as large-scale axes or unusual asymmetries, should be tested for their relation to superstructures, voids, and filament networks.
- If the CMB represents a coupling band of the field, it should not be completely independent of later large-scale field organization.
- The homogeneity of the CMB should be understood as an indication of a strongly smoothed background organization of the field, while its anisotropies could indicate local and large-scale residual couplings.

Black Holes, Anektron, and Anun

- Supermassive black holes should not appear as qualitatively more singular objects, but as larger saturation regions of the same boundary principle.
- Event horizons should be readable not as absolute breaking points of physics, but as regime boundaries between freely propagating and internally bound field motion.
- Jets and boundary-field reliefs should arise preferentially where saturation zones can no longer bind excess field tension further locally.
- Active galactic nuclei should represent especially strong reorganization zones of the field and therefore show spectral, lensing, and structural consequences.
- The environment of black holes should be characterized not only by mass, but also by saturation, relief channels, magnetic order, and field coupling.
- Growing black holes should not only bind their environment gravitationally, but increasingly influence large-scale structural organization as regime nodes.
- The Anun of one regime should be able to assume Anektron function in a superregime, which would have to show itself in scale-relative structural transitions and new coupling spaces.

Matter Formation, Inertia, and Magnetism

- Matter should be readable as bound field motion whose internal dynamics does not become slower, but is transferred into stable circulations, bindings, and reorganization patterns.
- Inertia should be connected to the internal binding architecture of a system: the stronger and more nested the internal organization, the greater the resistance to forced reorganization.
- Magnetism should occur especially strongly where field motion develops directed twisting, coherence, and near-order.
- Magnetic order should be examined together with rotation axes, plasma flows, charge asymmetries, and large-scale field tension.
- In stars, planets, accretion disks, and spiral arms, magnetic fields should not be mere accompanying phenomena, but should provide indications of directed twisting orders of the field.
- Differences between magnetic materials should, within the UQSH framework, be traceable to different capacities for coherent field twisting and coupling to larger field orders.
- Divided magnets should form their polarity again because the twisting order does not arise as an external addition, but as a local organizational form of the bound field structure.

Chemistry, Life, and Complex Organization

- Chemistry should be describable as flexible coupling of bound field nodes, not only as an abstract exchange of electrons.
- Chemical bonds should be readable as stable field bridges between atomic field organizations.
- Molecules should be especially stable where local tension, charge, geometry, and coupling conditions allow a repeatable field bridge.
- Water should therefore play a special role within the UQSH framework and be very common.
- The simplest and most frequently recurring elements should be hydrogen and helium. Within the UQSH framework, they would be the most natural stable basic forms of early or newly released matter organization. Especially in the vicinity of active galactic nuclei, jets, and other matter-ejection or reorganization zones, accumulations of hydrogen and helium gas should therefore occur frequently. Such gas regions would not be merely residual material, but indications that highly saturated field nodes can return matter to simple, stable coupling forms or eject it.
- Life-capable environments should be repeatedly possible when water, energy flow, chemical coupling spaces, and temperature and pressure windows coincide.
- Life should be investigable not as a random exception, but as a possible higher form of stable, open field organization in suitable coupling windows.

Scale Regimes, Planck Limit, and Observability

- The Planck limit should not necessarily appear as the final structural boundary of the field, but as a visibility and resolution boundary of our regime.
- Below the Planck limit, field motions should not disappear, but only no longer be directly resolvable in the terms of our regime.
- Subregimes and superregimes should not be understood as separate worlds, but as neighboring organizational layers of the same field reality, and should be tested indirectly through coupling effects.
- Systems of different size should show similar organizational patterns if they assume comparable roles in their respective regime.
- Observability should always be tied to coupling capacity. Not every existing field organization has to be directly visible.
- Non-coupling field organization should be detectable indirectly through gravitation, lensing, dynamics, or spectral modulation.

Speed of Light, Time, and Causality

- The speed of light should remain preserved as the maximum reorganization rate of free field motion, and need not be interpreted as the velocity of a transported object. In strongly tensioned or gravitational field regions, an external observer should perceive the transmission of light and processes as time-dilated, so that they may appear slowed. Locally within the tensioned region, however, a co-moving observer should measure neither a change in the speed of light nor an immediate change in their own time rate. The deviation therefore does not concern the local value of c , but the different timing and readability of bound field organizations between differently tensioned regimes.
- Within the UQSH framework, time dilation should be connected to altered timing of bound field organization, not to a reduction of c .
- Bound systems should show different reorganization rates depending on field tension and degree of organization, while the limiting rate of free field motion remains invariant.
- Extreme gravitational systems should not show that light becomes slower, but that the coupling and transmission of free field motion appear increasingly time-dilated from the perspective of external observers.
- In data, time should appear as physically different timing wherever field motion is more strongly bound, denser, or organized closer to saturation.

Overall Testing Strategy

- UQSH predictions should preferably be tested where several observational levels coincide: spectrum, rotation, lensing, gas, magnetic fields, CMB, redshift, and large-scale structural environment.

- Individual data levels should not be evaluated in isolation if the UQSH is correct, since field organization can express itself electromagnetically, dynamically, gravitationally, and structurally at the same time.
- The strongest tests therefore lie in combined maps: spectral modulation plus lensing, rotation plus gas structure, redshift plus void path, CMB plus large-scale structure, magnetic fields plus plasma flow.
- The UQSH would be weakened if it were shown that dynamic excess, lensing, redshift, CMB anisotropies, and spectral modulation can be explained completely independently of field organization, coupling paths, and baryonic structure.
- It would be strengthened if recurring organizational roles remain detectable across different scales and established measured quantities can be brought together within a shared field framework.

Experimental Test of the Wave–Particle Dualism

Within the UQSH framework, the wave–particle dualism can be tested through an extended double-slit or single-photon experiment. The central question is not whether light shows wave-like and point-like properties. That observation is well established. Rather, the test should ask whether the behavior can be understood as a combination of extended field motion and local coupling.

Within the UQSH framework, a light excitation is not understood as a small object flying through a slit. It is interpreted as free, spherically propagating field motion of Φ . Detection appears point-like because the extended field motion couples only locally to a bound field node of the detector. The wave-like aspect therefore belongs to the propagation of the front, while the particle-like aspect belongs to the local measurement coupling.

A suitable experiment would therefore have to examine three levels at the same time:

- the spatial propagation of the light front,
- the internal phase, frequency, and coherence structure of the front,
- the local coupling to detector matter.

An idealized setup would consist of an as point-like as possible single-photon source, or a source of ultrashort light pulses, a double-slit or multi-slit arrangement, and a detector plane with high spatial and temporal resolution. In addition, a weak reference or test pulse could be used to examine whether the passage of the light front leaves behind a short-lived change in the local coupling conditions.

Schematically, this gives:

$$\boxed{\text{Emission} \rightarrow \text{spherical field front} \rightarrow \text{interference structure} \rightarrow \text{local detection}} \quad (27)$$

In the standard picture, the interference pattern is described as a probability distribution for individual photon events. The UQSH adopts this observable structure, but reinterprets it field-mechanically: it is not a point-like particle that interferes with itself, but an extended field front that passes through both

slits as a coherent motion structure. The individual point on the screen is not the light itself, but the local coupling event between the front and a bound state of the detector.

A first test quantity is therefore the stability of the interference pattern at strongly reduced intensity. If the source is attenuated so that the detection events occur individually, the pattern should still build up statistically:

$$N(x) \propto I_{\text{front}}(x), \quad (28)$$

where $N(x)$ denotes the frequency of local detection events and $I_{\text{front}}(x)$ describes the intensity structure of the extended field front at position x . Within the UQSH framework, this means that the point structure of the measurement arises locally, while the underlying propagation remains front-like.

A second test quantity concerns the question of whether detection itself depends on the coupling capacity of the screen. Different detector materials, temperatures, field environments, or polarization filters should not only change the number of hits, but also provide indications of how the extended field front couples locally into bound states. Formally, this can be written as a coupling filter:

$$N(x) \propto \mathcal{F} * \det [I * \text{front}(x), M_{\text{front}}(x), K_{\text{det}}(x)] . \quad (29)$$

Here, $\mathcal{F} * \det$ denotes the detector coupling filter, $I * \text{front}(x)$ the local intensity of the front, $M_{\text{front}}(x)$ its modulation structure, and $K_{\text{det}}(x)$ the local coupling state of the detector material.

A third and more UQSH-specific test quantity would be a possible after-effect of the front on a subsequent test pulse. For this purpose, a strong, short pump pulse could be sent through the arrangement, followed by a weak probe pulse. The search would be for a very small, short-lived change in phase, travel time, polarization, or coherence of the probe pulse:

$$\Delta\varphi_{\text{probe}} = \varphi_{\text{probe}}^{\text{after}} - \varphi_{\text{probe}}^{\text{before}} . \quad (30)$$

Such a measurement would be relevant for the UQSH only if it cannot be explained by known effects such as heating, scattering, material response, nonlinear optics, or electromagnetic noise. The decisive test would be whether a propagating light front produces a measurable, spatially ordered, and short-lived change in the coupling conditions.

The experimental core can therefore be summarized as follows:

Wave	=	extended field motion
Particle	=	local coupling event
Dualism	=	regime transition between propagation and detection

(31)

The UQSH therefore does not predict that the known double-slit experiment must produce different hit patterns. Rather, it predicts that the known hit patterns can be understood as local coupling of a truly extended field front. This interpretation becomes testable above all where propagation, interference, detector coupling, and possible after-effects of the same front are measured together.

25.1. Prediction on the Origin of Life in the Universe

The UQSH leads to a clear qualitative prediction regarding the origin of life: life should not be a one-time special case in the universe, but should become repeatedly possible wherever stable coupling windows of matter, energy flow, water chemistry, and sufficient time arise. This statement does not mean that life must automatically be present on every suitable celestial body. It does mean, however, that under comparable conditions the field medium should tend repeatedly toward similar organizational chains.

The reason lies in the same working mode of the Qu-foam across different scales. Field motion forms stable nodes, these nodes make matter possible, matter makes chemistry possible, and chemistry can, under suitable conditions, pass into self-reorganizing structures. Life therefore does not appear as a principle added from outside, but as a possible continuation of the same field logic: stability, coupling, energy flow, uncertainty, reorganization, and adaptation.

Water plays a special role because it provides a broad chemical coupling space. If hydrogen and oxygen are available, if suitable temperature and pressure windows exist, and if energy sources such as starlight, geothermal activity, chemical gradients, tidal forces, or radiation fields are present, the transition toward complex chemistry should be repeatedly favored. In this view, water is not magical, but an especially effective medium for flexible field coupling, bond release, charge separation, heat transport, and molecular self-organization.

The UQSH therefore predicts that life-friendly environments should not be sought only on Earth-like planets. Subsurface oceans, icy moons, hydrothermal systems, water-rich dwarf planets, dense atmospheres, interfaces between ice, rock, and water, as well as chemically active boundary zones of planets and moons, can also form suitable coupling windows. What matters is not external similarity to Earth, but the long-term stability of a chemical reorganization space.

A second prediction follows from this: life forms should develop differently depending on their local environment. In water-rich oceans, different structures would be expected than in icy worlds, hydrothermal boundary regions, dense atmospheres, or strongly radiation-loaded environments. Life would then not be a fixed form, but purpose-bound self-organization within a local field and chemistry window. Its form follows the environment, because every living being gains its stability from the energy, water, temperature, matter, and coupling conditions present there.

This prediction becomes testable through the frequency and diversity of prebiotic chemistry. If the UQSH interpretation is correct, complex organic molecules, water chemistry, stable chemical gradients, and possible precursors of biological self-organization should occur in many cosmic locations, not only within a narrow Earth-like range. Especially meaningful would be recurring patterns of similar chemical organization in very different environments, such as protoplanetary disks, comets, ocean-bearing moons, the Martian subsurface, exoplanet atmospheres, or hydrothermal systems.

The UQSH expects life not as a one-time accident, but as a recurring possibility of stable field, water, energy, and chemical coupling.
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If the universe is not reduced to a short, one-time window of emergence, but is understood as a spatially, temporally, and scale-relatively much more open field process, then the possible time window

for life also expands. Life forms may then not only have arisen once, but may arise, persist, disappear, and reappear under new conditions again and again.

This leads to an extended prediction: life forms can exist in the universe that are comparable to our development, that appear much younger or more simply organized, or that are older, more stable, and more highly developed. Some lines of life may already have passed, others may only just be emerging, and still others may have adapted over very long periods to stable coupling windows. In this view, life is not an exception against the field, but an expected organizational form wherever matter, energy flow, water chemistry, duration, and reorganizational capacity sufficiently come together.

Consciousness can be described as a higher-order form of self-maintaining organization. It would then not be understood as a substance separate from the field, but as the internal feedback capacity of a living system: the system not only preserves its structure, but forms models of its environment, its boundaries, and its possible actions. Consciousness thus appears as the self-reference of an organization that dynamically co-regulates its own persistence, its couplings, and its future possibilities.

This view does not claim that every life form possesses consciousness. Rather, it classifies consciousness as a possible higher degree of organization: life stabilizes itself; conscious life recognizes and steers parts of this self-stabilization. Consciousness thereby also remains embedded in the general logic of the UQSH: field motion becomes matter, matter becomes chemistry, chemistry becomes life, and life can, under suitable conditions, become self-referential, future-oriented organization.

More highly developed life forms can then be understood as systems that not only exist within the Qu-foam, but increasingly recognize and use its coupling conditions. Such development would not necessarily have to become louder, larger, or more energetically conspicuous. On the contrary: the better a civilization understands the working mode of the field, the more likely it could act technologically in symbiosis with the Qu-foam.

Technological symbiosis here means that one does not work against the field, but with its pathways of reorganization. Locomotion would then not primarily be thrust against an environment, but controlled coupling to local and non-local field states. Energy generation would not merely be combustion or coarse release of stored matter, but targeted relief, densification, or reorganization of existing field tension. Matter generation would not be creation from nothing, but the controlled transfer of free field motion into bound stable field nodes.

A very highly developed life form could therefore appear quiet, but effective, on a cosmic scale. Its technology would not have to produce strong waste traces, massive thermal signatures, or violent energy excesses if it couples, redirects, and reintegrates field tension efficiently. It would act less like a consumer of the universe and more like a local co-organizer of the field: stabilizing, relieving, shifting, densifying, or connecting.

From this perspective, advanced civilizations would not necessarily be recognizable through ever larger machines or stronger radiation output. It is also possible that they would leave signatures of unusually coherent field order: controlled relief zones, apparently low-loss motion, abrupt changes of direction without classical propulsion logic, local changes in inertial responses, artificially stabilized matter forms, or structures that organize energy and coupling flows with striking efficiency. Such considerations remain speculative, but follow from the UQSH assumption that matter, motion, gravitation, and energy are different organizational forms of the same Qu-foam.

This broadens the question of extraterrestrial life. The search would not only be for biological traces or strong technological emissions, but also for signs of field-symbiotic organization: patterns in

which energy flow, matter generation, motion, and gravitation appear unusually coordinated. A highly developed life form would then perhaps not be the loudest appearance in the universe, but the one that works most quietly and precisely with its fundamental motion.

26. Outlook: Field Coupling, Location, and Non-Classical Motion

If space and time, within the UQSH framework, are not absolute containers but regime-dependent readings of the field medium, then a theoretical outlook opens onto a different kind of motion. Motion would not necessarily have to mean that an object travels a distance through an already given space. It could also mean that a stable field node changes its coupling to the surrounding field order.

In the classical picture, a body is located at one place and must travel through space to reach another place. Within the UQSH framework, however, a location is not an absolute position in an empty container, but a local coupling state of the field. A location is the way in which a field region is readable, stabilized, and connected relative to other field regions. Motion is therefore not only a change of location, but a reorganization of field binding.

This gives rise to a speculative possibility: if a system could influence the coupling structure of the field medium deeply enough, it would not have to traverse the entire distance classically. Instead, it could attempt to modulate the local field order so that it becomes resonant with the coupling signature of a distant target region. The target location would then not be “brought closer” in the ordinary sense; rather, its field signature would be made locally connectable.

classical motion = traversing a distance,

field-mechanical motion = reorganization of coupling to field states.

In this view, distance would not be an absolute barrier, but a regime-dependent coupling difference. An advanced field technology would therefore not have to exceed the speed of light, but would have to change the local coupling to the deeper field order.

An intuitive formulation would be this: one does not simply imitate the matter of a target location, but the field conditions under which that location appears as a location. If the local environment of a system assumes the same coupling signature, the same tension order, the same phase structure, and the same reorganization condition as a distant field region, a kind of field-mechanical connection could arise. The target region would then not be reached by traversing a distance, but through resonance with its field order.

target location $\sim$ coupling signature of the field.

travel $\sim$ transition between coupling signatures.

For this to happen, several conditions would have to be fulfilled simultaneously. First, the local field tension would have to be controllably alterable. Second, the system’s own matter form would have to remain stable even though its coupling to the surrounding space state is changed. Third,

the target signature would have to be generated not merely approximately, but in a phase- and structure-compatible way. Fourth, the process must not represent a violation of the local limiting rate c . The local reorganization of the field would remain bound to c ; what would change is not the speed of light, but the assignment of the system to a spatial field reading.

Such motion would therefore not be classical faster-than-light flight. It would rather be a regime transition of location coupling. The object would not move faster through space; instead, it would change the spatial relation in which its position is defined.

faster-than-light flight $\neq$ field-coupling transition.

location = stabilized field relation.

non-classical motion = controlled change of this field relation.

With regard to reports of unusual flying objects or abrupt movements, such a UQSH interpretation would be only a speculative interpretive possibility, not evidence. If such phenomena were real, technological, and not misinterpreted, an explanation would probably not begin with classical propulsion, but with coupling, inertia modulation, and field geometry. Sudden changes of direction, apparent absence of inertia, or extremely rapid changes of location would then not be expressions of unlimited acceleration of a body, but indications that the body's coupling to the local field order is being changed.

The decisive point is inertia. In the normal regime, a body cannot be accelerated arbitrarily, because its bound field organization resists reorganization. If, however, the coupling to the surrounding field geometry is changed, the required reorganization work could decrease. The body would not be accelerated against its inertia, but brought into a field state in which the inertial response is organized differently.

Inertia = resistance of bound field organization to reorganization.

inertia modulation = change in the coupling conditions of this organization.

The popular idea that one would have to copy or imitate the “vibration” of a place can be interpreted in UQSH language as a coupling signature. What would be meant is then not a mystical vibration, but a very concrete bundle of frequency structure, field tension, phase, curvature, saturation, local timing, and coupling geometry. A place would not be only a coordinate, but a field state. Whoever could reproduce this state in a controlled way would not copy the location as matter, but establish connection to its field relation.

This gives the speculative core statement:

It is not the distance that would have to be overcome, but the coupling difference between two field locations.
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Such an outlook remains far beyond the present technical and experimental state. Within the UQSH,

however, it is logically connectable: if space and time are readings of the field, then the deepest form of locomotion would not be propulsion through space, but the controlled change of the field readability of location and time.

27. Comparison with Established Models

The UQSH does not replace general relativity. It reads its curvature of spacetime as the expression of physically carried field tension. General relativity remains the precise geometric description; the UQSH attempts to interpret the physical carrier structure of this geometry.

The standard model of cosmology is likewise not simply rejected. The CMB, redshift, lensing, structure formation, and the distribution of galaxies remain real findings. The UQSH, however, does not interpret them as the necessary consequence of the expansion of a passive space from a singularity, but as the result of ongoing field reorganization, dedensification, and coupling thresholds.

Compared with particle models of dark matter, the UQSH avoids the assumption of an additional invisible substance. Dark-matter effects are interpreted as organized, saturation-dependent field tension. Compared with MOND [23], the UQSH shares the importance of baryonically coupled dynamics, but interprets the relation as field organization rather than as a mere modification of an acceleration law.

Related emergent interpretations of gravitation include thermodynamic and entropic approaches [20–22]. The UQSH shares with them the idea that gravitation need not be an isolated fundamental substance, but differs through its specific carrier interpretation: gravitation, light, spectrum, CMB, black holes, the Planck limit, and the dark-matter effect are described together through a dynamic field medium with tension fronts, saturation, and coupling filters.

28. Possible Misunderstandings and Limits

The view developed here should not be understood as ignoring established measurements or mathematical models. General relativity remains an extremely successful description of gravitational geometry. The UQSH attempts to interpret this geometry field-mechanically. Likewise, spectral analysis, redshift, lensing, and the CMB remain real observational findings. The question is not whether these findings exist, but how they are read ontologically.

This work likewise does not claim that electromagnetic spectra alone contain the complete field configuration of a galaxy. They are only one layer. The actual field organization must be inferred through several projections: spectrum, rotation curves, baryonic structure, gas distribution, lensing, CMB environment, large-scale embedding, and dynamic activation.

The galaxy–atom analogy is also not an identification. The analogy concerns the principle: bound organization can characteristically modulate propagating fronts. Atoms do this in the electromagnetic regime; galaxies could do so in the superregime through gravitational and electromagnetic field organization.

The data analyses carried out so far should also be read as first indications. SPARC- q regimes show structural differences, while simple WISE/GALEX/SDSS colors do not provide a robust separation. The template smoothing analysis supports the idea of coarse spectral organizational forms, but is limited by the small number of templates. Larger, homogeneous datasets are required for a robust test.

29. Conclusion

The present work develops *gravitational light* as a scale-relative UQSH interpretation of radiation, matter, gravitation, spectrum, and observation. Its starting point is the assumption that the Qu-foam, or field medium, is not merely the background or stage of physical processes, but the shared dynamic ground structure in which all observable states arise, couple, relax, and reorganize.

The central unification lies in treating radiation, matter, and gravitation not as ontologically separate entities, but as different organizational forms of the same field motion. Radiation is the free, propagating form of this motion. Matter is its bound and stabilized form. Gravitation arises wherever field curvature and field tension occur. It is therefore not only the persistent geometry of bound matter, but the regime-dependent response of the field to every organized curvature tension. Its effect is nevertheless limited within each regime: no regime can translate field tension into local depth without limit.

Radiation = free tension front,

Matter = bound field tension,

Gravitation = response to organized field curvature and field tension.

This shifts the physical question. The focus is no longer on individual objects, but on the way field motion becomes bound, released, dedensified, densified, modulated, or observable. Light appears as a free tension front whose internal modulation is readable as spectrum. What we see as light is not the complete field front, but a regime-compatible modulation. What we measure as a spectrum is not only a frequency distribution, but the readable organizational structure of a carrying tension front.

Matter appears as a stabilized coupling form. Gravitation appears as a regime-limited field response to curvature, tension, and organization. Observation itself becomes a regime-dependent coupling between an incoming field order and the material structure of the observer. An observer therefore never reads the complete field reality, but only the layer that can interact resonantly with their regime, their matter form, and their coupling filter.

Within this framework, the dark-matter and dark-energy effects also receive a shared reading. The dark-matter effect appears as baryonically anchored, radiation-excited, and saturation-limited field tension. Dark energy appears as a local relaxation tendency of the field that becomes especially prominent in voids. Cosmological redshift is not disputed, but interpreted as the dedensification of propagating tension fronts.

This leads to a unified interpretation of light, spectrum, dedensification, the CMB, galaxies, q -regimes, the dark-matter effect, the dark-energy effect, black holes, the Planck limit, quantum numbers, and interactions. The UQSH does not claim that all details have already been fully formalized. It does, however, offer a shared language in which many separate problems appear as different readings of the same field organization.

Anektron and Anun mark the lower and upper saturation boundaries of a regime. The Anektron stands for minimal stable coupling capacity, the Anun for maximum local field saturation. The decisive recursive idea is this: the Anun of one regime can assume Anektron function in the next higher regime.

Saturation is therefore not the end of field motion, but a transition mechanism.

The uncertainty that arises in this process is not a weakness of the model, but a condition of stability. At the lower boundary, it protects the minimal coupling form from absolute fixation; at the upper boundary, it prevents singularity and enables redistribution, growth, and regime connection.

The UQSH understands itself as a field-ontological interpretive level: known phenomena should not stand separately beside one another, but should appear as different readability states of a single dynamic field medium.

Explanation does not take the magic away from the universe. It only changes where that magic resides. The wonder no longer lies only in what is not yet understood, but in the depth of connection. The more order becomes visible, the greater the astonishment becomes. For then it becomes clear: the world is not carried by chaos, but by a precision that reaches into every tension, every modulation, and every form.

Energy is field motion; radiation, matter, and gravitation
are its different organizational forms.

Magic becomes greater through understanding.

Final Remark on the Qu-Ground

At the end of this work, one question remains open that cannot be answered fully by formulas: What actually is the Qu-ground Q ?

This work has primarily described physical processes. It dealt with light, energy, gravitation, matter, inertia, magnetism, time, redshift, and dark effects. All these phenomena were interpreted as readings of a deeper field motion. With Q , however, what is meant is not another field within this physics. Q denotes the ground from which a physical field reality can arise in the first place.

This makes the concept difficult to grasp. We are used to connecting reality with something tangible: with bodies, places, measured values, motion in space, or change in time. But Q lies precisely before this level. For this reason, Q has in this work no definable zero, no definable unit, and no location in space and time. A zero, a one, a scale, or a measured value already belong to a world in which something has become distinguishable and measurable. Q is not this measurable state, but that from which measurability first becomes possible at all.

The physically accessible level is therefore denoted by Φ . By Φ , what is meant is not an illusory world, not a virtual environment, and not a projection in the sense of a projector image. Φ is the real field reality in which Q appears to a regime as space, motion, tension, curvature, coupling, and organization.

A direct measurement link between Q and Φ is not claimed here. According to the present state of this work, it is not even fully explainable how Q and Φ could be distinguished according to the physical standards of our universe. For every measurement already presupposes space, time, coupling, and observation. It therefore already takes place within Φ .

Precisely for this reason, Q should not be too quickly understood as a physical object. It is also not a substance, not a force, and not empty space. Q is a boundary concept. It marks the point at which physical description points back to its own origin, without being able to measure that origin itself

completely.

Perhaps a simple image may help, without taking it literally. Thoughts can take infinitely many forms. In thought, something can arise, change, disappear, return, and be ordered anew, without the thought itself being a spatial object. One can imagine worlds within it without this imagination being bound to a place, a distance, or a measurable boundary. In a similar way, Q can be thought of as an intangible ground from which physical forms first emerge.

This does not mean that Q is consciousness in the human sense. It means only that our ordinary images of space, body, and measurability are no longer sufficient for this ground.

At this point, the UQSH also touches terms that have traditionally been used more philosophically or religiously: existence, origin, wholeness, consciousness, or God. These terms are not introduced here as physical definitions. At most, they can indicate why Q should not be understood as a single something within the world. If everything that appears physically arises from this ground, then Q is not a part of the whole, but the prerequisite for a whole to appear at all.

The question of the beginning and end of the universe also remains open thereby. If cosmological redshift does not arise mainly through an expansion of space, but through the dedensification of propagating field motion, then the visible boundary of the universe would not be an absolute boundary of reality. It would only be the boundary of what our regime can still couple to and measure. The universe could then be older, larger, and more open than it appears within a purely expansion-based picture.

The UQSH therefore does not claim to explain the Qu-ground definitively. It only proposes not to equate measurable physical reality with its deepest ground. What remains formally accessible at first is Φ , the field reality. Q remains the prior ground that cannot be measured directly, but becomes indirectly visible in the physical readability of Φ .

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